CTR
Vertex Shader Reference Manual

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Version 4.2

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1 About This Document

This documentation describes vertex shaders that run on the CTR-SDK.

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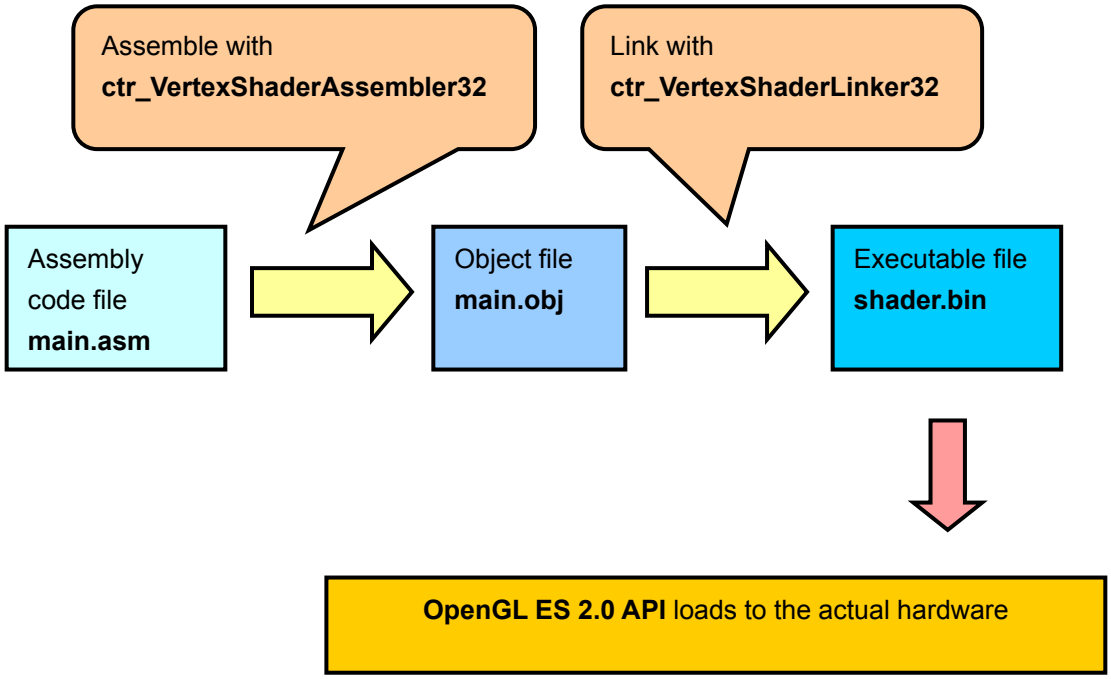
2 Overview

Vertex shaders are the only programmable shaders implemented by the CTR-SDK. Vertex shaders are written in PICA-specific assembly language. To generate executable files, the `ctr_VertexShaderAssembler32.exe` and `ctr_VertexShaderLinker32.exe` programs assemble and link vertex shaders written in this assembly language.

The OpenGL ES 2.0 API loads the executable files and then runs the shaders. The OpenGL ES 2.0 API can load only executable files that have been assembled and linked in this way. You cannot load shader files written in GLSL.

Figure 2-1 Vertex Shader Overview



3 Operating Environment

The `ctr_VertexShaderAssembler32.exe` and `ctr_VertexShaderLinker32.exe` programs have been confirmed to run in the following operating environment.

Table 3-1 Operating Environment Specifications

Component	System Requirements
CPU	Pentium 4 3.06 GHz
Memory	1 GB
OS	Windows XP
Development Environment	Microsoft Visual Studio .NET 2003

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4 Main Objects and Reference Objects

Vertex shaders are run from the `main` function. A *main object* is an object file that can be assembled from an assembly code file that has a `main` function. A *reference object* is an object file that can be assembled from an assembly code file that does not have a `main` function.

To put a `main` function in an assembly code file, set the `main` label at the location where shader execution starts and set the `endmain` label after the last instruction at the end of the `main` function. (This is the last instruction in the `main` function, not the last instruction in the assembly code file.)

A reference object only has subroutines and is referenced by a main object to resolve unresolved labels. When the objects are linked to create an executable file, the executable file does not include reference objects that are not referenced by any main objects.

If multiple main and reference objects are linked, a single executable file is generated that includes more than one main object. Specify the executable file with the `glShaderBinary` function to load it. The number of shader objects you specify to the `glShaderBinary` function at that time specifies the number of main objects to load.

Main objects do not reference each other for unresolved labels when they are linked. Also note that if multiple reference objects with the same label name are specified as link targets, an error occurs while linking.

You cannot link only reference objects to create an executable file; you must link at least one main object.

5 How to Use the Assembler Tools

5.1 ctr_VertexShaderAssembler32 (Assembler)

`ctr_VertexShaderAssembler32.exe` assembles assembly code files written in a PICA-specific assembly language and outputs object files.

The assembler is run from the command prompt as follows.

```
ctr_VertexShaderAssembler32 <input filename> [options]
```

Specify the input assembly code file in `<input filename>`. You can specify the options in section 5.1.1 for `[options]`.

This command has the following characteristics and requirements.

- An input file must be specified.
- The options can be omitted.
- Help is displayed when no arguments are specified.
- The Shift-JIS and UTF-8 (with a byte order mark) file encodings are supported.
- The CR+LF newline code is supported.
- Use a filename that does not contain any spaces, is composed of no more than 128 single-byte alphanumeric characters, and does not use the symbols `\ / : * ? " < > |`.

5.1.1 Options

- `-O <filename>`
  Specifies the output filename. (This is `<input filename>.obj` if it is left unspecified.)

- `-I <file path>`
  Specifies the input file path. You can use this option to specify the path to both the assembly code file and the files that it includes. First the current directory and then the directory specified by this option are searched for the input file and include files.

- `-D <key> [= <value>]`
  Defines a macro. Specify the macro name with `<key>` and its value with `<value>`.The `<value>` can be omitted.

- `-debug`
  Generates an object file with debugging information. When this option is not specified, the output object file does not include the full input file path.

- `-nowarning`
  Suppresses warning message output.

- `－preprocess`
  Runs preprocessing only. Runs preprocessing on the input assembler file, and outputs the result after deleting characters that have no effect (such as spaces and comments) to standard output.

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File information is added to the output results in the format "*# line-number filename*". Expands assembler macro instructions, and replaces the instructions with the expansions. However, if there is a syntax error in the macro instruction, the instruction is not replaced by the expansion.

- `-dependency`
  Outputs the assembler file's dependencies. Only preprocesses the input assembler file, then outputs the input filenames and all of the included filenames referenced in the input file to standard output. The `-preprocess` option is ignored if specified together with `-dependency`.
- `-?` or `-help`
  Displays Help.

5.1.2 Example

Code 5-1 ctr_VertexShaderAssembler Example

```
ctr_VertexShaderAssembler32 main0.asm -IC:/sample/src -IC:/sample/inc -DDEBUG=1
```

In this example, `main0.asm` is the name of the input file; `C:/sample/src` and `C:/sample/inc` configure the input file paths. The `DEBUG` macro is also defined with a value of 1.

When this input file is successfully assembled, the file `main0.obj` is generated.

5.2 ctr_VertexShaderLinker32 (Linker)

`ctr_VertexShaderLinker32.exe` links object files output by `ctr_VertexShaderAssembler32.exe` and then outputs an executable file.

The linker is run from the command prompt as follows.

```
ctr_VertexShaderLinker32 <input files> [options]
```

Specify the input object files in `<input files>`. The input files must be object files that together include at least one `main` function. You can specify (or omit) the options in section 5.2.2 for `[options]`. Help is displayed if no arguments are specified. Use filenames that do not contain any spaces, are composed of no more than 128 single-byte alphanumeric characters, and do not use the symbols `\ / : * ? " < > |`.

If there is more than one main object in the input files, the application must specify the same number of shader objects to the `glShaderBinary` function. At that time, each shader object references a main object in the order the main objects were specified in the argument to `ctr_VertexShaderLinker32.exe`.

5.2.1 Input files

- `<input filenames> [input files]`

The input files must be object files that together include at least one `main` function.

5.2.2 Options

- `-I <file path>`
  Specifies the input file path. First the current directory and then the directory specified by this option are searched for the input files.
- `-O <filename>`
  Specifies the name of the output file. This is `shader.bin` by default.
- `-M`
  Outputs a map file. This file has the same name as the executable file, but uses the `.map` extension. For more information about map files, see Chapter 11 Map Files.
- `-debug`
  Links all linked object files with debugging information.
- `-nodebug`
  Links all linked object files without debugging information.
- `-warn_level=<level>`
  Sets the level of the warning, as specified by <level>. For more information, see Chapter 14 Warning Levels.
- `- check_performance`
  Conducts a performance check on all linked main objects. For more information, see section 15.2 Performance Checker Feature.
- `-?` or `-help`
  Displays Help.
- `-check_consistency`
  Performs a consistency check of all linked main objects. This option is retained for compatibility. In the current specification, the running of the consistency check is based on the warning level that has been set. For more information, see Section 16.1 Consistency Checker Feature and Chapter 14 Warning Levels.

5.2.3 Example

Code 5-2 ctr_VertexShaderLinker Example

```
ctr_VertexShaderLinker32 main0.obj main1.obj subr0.obj subr1.obj —Oshader0.bin —M
```

This example links the main objects `main0.obj` and `main1.obj`, and the reference objects `subr0.obj` and `subr1.obj`.

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6 Vertex Shader Resources

Vertex shaders have the following resources.

6.1 Program RAM

Program RAM is the region that stores assembly language instruction codes. It can store 512 instructions. If an assembly code file uses more than 512 assembly instructions, it causes an error when it is assembled or linked.

6.2 Registers

6.2.1 Per-Register Resources

The following register types are used for calculations and flow control.

Table 6-1 Register Types

Name	ID	Components	Number	R/W	Index	Bit Width
Input registers	`v#`	4	16	R	-	24
Temporary registers	`r#`	4	16	RW	-	24
Floating-point constant registers	`c#`	4	96	R	`a0/aL`	24
Address register	`a0`	2	1	RW	-	8
Boolean registers	`b#`	1	16*	R	-	1
Integer registers	`i#`	1	4	R	-	24
Loop-counter register	`aL`	1	1	R	-	8
Output registers	`o#`	4	16	W	-	24
Status registers	–	1	2	RW	-	1

Note: Boolean register 15 (b15) is reserved for the geometry shader.

- ID
  This identifier is used when entering assembly instructions. "#" indicates the register number. Specify a number from 0 to (number of registers - 1).
- Name
  This is the name of the register.
- Number
  This is the number of registers.
- Components
  This is the amount of data in a single register. When there are four components, a single register contains the x, y, z, and w components.

- R/W
  This indicates whether reads and writes are allowed. "R" indicates that a register can be specified only as an input operand. "W" indicates that a register can be specified only as an output operand. "RW" indicates that a register can be specified as either an input or output operand.
- Index
  You can specify the register numbers of these registers by using the content of other registers in Table 6-1. For more information, see section 7.5 Using Register Index Offsets for Input Operands.
- Bit Width
  This indicates the bit width of each register.

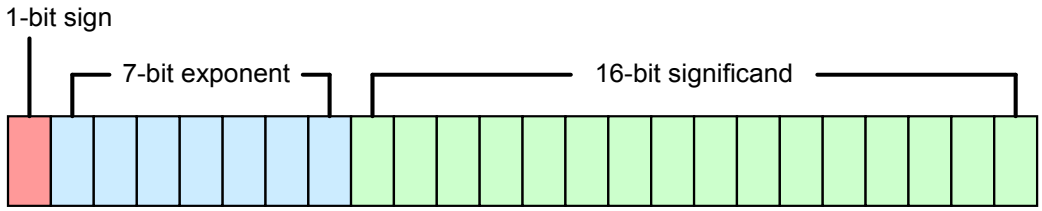
6.2.2 Precision of Floating-Point Registers

The input registers, temporary registers, and floating-point constant registers are all floating-point number registers. Floating-point numbers use 1 sign bit, 7 exponent bits, and 16 significand bits for a total of 24 bits. A sign bit of 0 is positive and 1 is negative. The exponent bits are in base 2 and have a bias of 63. The significand bits represent a value that is one less than the actual significand.

The actual value of a floating-point number is:

$$(-1)^{(\text{sign})} \times 2^{(\text{exponent}-63)} \times (1 + \text{significand})$$

Figure 6-1 Floating-Point Number



6.2.3 Input Registers

Input registers are floating-point registers that store vertex attribute data ("attributes" in OpenGL ES 2.0 applications).

6.2.4 Temporary Registers

Temporary registers are floating-point registers that temporarily maintain calculation results to be reused later. Their values are preserved until they are overwritten.

6.2.5 Floating-Point Constant Registers

Floating-point constant registers are floating-point registers that store constants to use for calculations. Uniforms for OpenGL ES 2.0 applications are stored here.

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6.2.6 Address Register

An address register is an integer register that accepts values in the range [-95, 95]. If you assign the value from a floating-point register to the address register, only the integer part is assigned. Operations are undefined if you assign values that are not in the range [-95, 95].

You can use this register's value to specify another register's number. For more information, see section 7.5 Using Register Index Offsets for Input Operands.

6.2.7 Boolean Registers

Boolean registers hold Boolean values. They are used for branches and jumps.

Uniforms for OpenGL ES 2.0 applications are also stored here. Boolean register 15 (`b15`) is reserved for geometry shaders.

6.2.8 Integer Registers

Integer registers hold integer values and are used to control `loop` instructions.

These registers store loop counts, the initial values for the loop-counter register, and the amounts by which to increment the loop-counter register. They are 24 bits wide: bits 0-7 specify the loop count; bits 8-15 specify the initial value for the loop-counter register; and bits 16-23 specify the amount by which to increment the loop-counter register.

When control enters a `loop` instruction, the loop-counter register is first initialized with its initial value from one of the integer registers. Then the assembly code instructions from the `loop` to the `endloop` instructions are run repeatedly. The number of loop iterations is the loop count plus one (meaning that the instructions are run only once when the loop count in this register is 0). At each iteration of the loop, the loop-counter register is incremented by the amount specified by this register.

Uniforms for OpenGL ES 2.0 applications are also stored here.

6.2.9 Loop-Counter Register

This register stores the counter value for loop instructions. Its value is in the range [0, 255].

You can use the value of this register to specify another register's number, just as you can with the address register. For more information, see section 7.5 Using Register Index Offsets for Input Operands.

6.2.10 Output Registers

These registers output data that has been processed by vertex shaders into a later stage of the graphics pipeline.

6.2.11 Status Registers

These registers have their values set by comparison instructions and are used for branch conditions.

7 Assembly Language Grammar Reference

7.1 Entering Assembly Instructions

Assembly instructions are entered in the following format. The comment can be omitted.

```
<Operation>  [Operand] [Comment]
```

Code 7-1 Assembly Instruction Example

```
add       r0, r1, r2      // r0 = r1 + r2
mul       r3, c0, v0      // r3 = c0 * v0
```

Assembly instructions are entered using ASCII characters, in compliance with the following rules.

7.1.1 Operation

For the operation, denote the assembly instruction to run. You can only code one operation per line.

7.1.2 Operand

```
Operand name[, Operand]
```

Denote the name of the operand, such as the register or the direct value targeted by the operation. The operation is followed by at least one space or tab before the operand name. Use commas to delimit multiple operands. You can denote one or more spaces or tabs between operands.

7.1.3 Comment

Denoting two forward-slashes ("//") causes the rest of the line to be treated as a comment. Text between the delimiters `/*` and `*/` is also treated as a comment.

7.2 Masking Output Components

When you output calculation results to registers that have more than one component, you can specify which components to output. Values are updated only for the specified components. If nothing is specified, all components are updated. Specify the components in x, y, z, w order. (You cannot, for example, denote "wzyx.")

Code 7-2 Component Masking Example

```
add   r0.x, r1, r2   // The x component is updated.
                   // The y, z, and w components are not updated.
mov   r0.yz, r1      // The y and z components are updated.
                   // The x and w components are not updated.
dp3   r0, r1, r2     // The x, y, z, and w components are all updated.
```


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7.3 Rearranging Input Components (Swizzling)

When using registers with multiple components as input operands, you can rearrange (swizzle) the components that are used by calculations. If nothing is specified, components are used in x, y, z, w order. If you do not specify all four components, the component specified last is repeated.

Code 7-3 Swizzling Example 1

```
add       r0, r1.xzyy, r2         // r0.x = r1.x + r2.x
                                  // r0.y = r1.z + r2.y
                                  // r0.z = r1.y + r2.z
                                  // r0.w = r1.y + r2.w
mov       r0, r1.ywz              // r0.x = r1.y
                                  // r0.y = r1.w
                                  // r0.z = r1.z
                                  // r0.w = r1.z ; The z-component, specified last,
                                  // is repeated
add       r0, r1.zw, r2.xy        // r0.x = r1.z + r2.x
                                  // r0.y = r1.w + r2.y
                                  // r0.z = r1.w + r2.y
                                  // r0.w = r1.w + r2.y
```

Note that input component swizzling and output component masking are interpreted differently when you do not specify all four components. The first, second, third, and fourth input components after swizzling are applied respectively to the x, y, z, and w components specified for output component masking.

Code 7-4 Swizzling Example 2

```
mov       r0.x, r1.xy             // r0.x = r1.x ; (r1.xyyy)
mov       r0.y, r1.xy             // r0.y = r1.y ; (r1.xyyy)
mov       r0.z, r1.xy             // r0.z = r1.y ; (r1.xyyy)
mov       r0.w, r1.xy             // r0.w = r1.y ; (r1.xyyy)
add       r0.zw, r1.yx, r2.wz     // r0.z = r1.x + r2.z;(r1.yxxx + r2.wzzz)
                                  // r0.w = r1.x + r2.z;(r1.yxxx + r2.wzzz)
```

As shown above, the input component marked in red is used for each output component.

7.4 Adding a Negative Sign to Input Components

You can prefix input operands with a negative sign.

Code 7-5 Negative Sign Example

```
add       r0, r1, -r2     // r0 = r1 - r2
mul       r0, -r1, r2     // r0 = (-r1) * r2
```

7.5 Using Register Index Offsets for Input Operands

You can offset the register number of an input operand by entering it in brackets ("[ ]"). You can enter the sum of multiple integers in the brackets. If this integer sum (indicating the offset) exceeds the number of registers, an error occurs during assembly.

The floating-point constant registers, and only these registers, can also be offset by the values of the address register and/or the loop-counter register. If you use the address register, you must specify either its x or y component. Operations are undefined when the total index offset, including the address register and loop-counter register, is not within the range [0, 95]

Code 7-6 Index Offset Example

```
c2               // The index is 2
v[5 + 2]          // The index is 5 + 2
r8[1 + 2 + 4]     // The index is 8 + 1 + 2 + 4
c[3 + a0.x]       // The index is 3 + a0.x
c4[11 + aL]       // The index is 4 + 11 + aL
```

7.6 Labels

Labels are specified as jump targets for the `call` and `jpb` instructions.

To encode a label, add a colon ("`:`") after a name that is a combination of single-byte alphanumeric characters and underscores ("_"). A decimal or hexadecimal number (such as `123` or `0xf`), a register name (such as r0 or c0), or any other reserved word cannot be used as a label. A line encoding a label cannot have any other notation except a comment.

Code 7-7 Label Example

```
function0:
  mov     r0, r1
  ...
  ret
main:
  ...
call      function0
```

7.6.1 The main Label

A vertex shader always begins executing at the `main` label. A shader without a `main` label is referenced as a subroutine. When vertex shader assembly code files are assembled and linked, at least one of the linked objects must be a main object that contains at least one `main` label. In the same way, main objects must also set at least one `endmain` label. Set the `endmain` label immediately after the last instruction to be run. (This is the instruction that does the final write to the output registers.)

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7.6.2 Label Name Collisions

You cannot use the same label name more than once in a single object file. Likewise, when linking multiple object files, you cannot use the same label name more than once in object files used for multiple subroutines. Main objects do not reference each other, so the same label name can be used within more than one main object.

7.7 Reserved Words

The assembler defines the following strings to be reserved words. Do not use reserved words for labels, symbols, or other names.

7.7.1 Register Names

`v0–v15`, `r0–r15`, `c0–c95`, `a0`, `b0–b15`, `i0–i3`, `aL`, `o0–o15`

7.7.2 Assembly Language Instructions

`def`, `defb`, `defi`, `add`, `dp3`, `dp4`, `dph`, `dst`, `exp`, `flr`, `litp`, `log`, `mad`, `max`, `min`, `mov`, `mova`, `mul`, `nop`, `rcp`, `rsq`, `sge`, `slt`, `sub`, `abs`, `crs`, `frc`, `lrp`, `m3x2`, `m3x3`, `m3x4`, `m4x3`, `m4x4`, `nrm`, `pow`, `sgn`, `sincos`, `call`, `callc`, `callb`, `jpb`, `jpc`, `ret`, `ifb`, `ifc`, `else`, `endif`, `loop`, `endloop`, `breakc`, `cmp`, `end`

7.7.3 Preprocessor

`include`, `define`, `undef`, `ifdef`, `ifndef`, `if`, `defined`, `else`, `elif`, `endif`, `error`, `pragma`, `bind_symbol`, `output_map`

8 Preprocessor Pseudo-Instructions

8.1 #include

Use this instruction to insert a file. Place the file in double quotes.

Code 8-1 #include Example

```
#include “defs.asm”
```

8.2 #define

This instruction defines a macro. A macro is denoted using no more than 128 single-byte alphanumeric characters and underscores ("_") and cannot begin with a number. Operations are undefined if more than 128 characters are used.

Code 8-2 #define Example

```
#define   MAX_COUNT              100
```

8.3 #undef

This instruction deletes a macro defined by `#define`.

Code 8-3 #undef Example

```
#undef    MAX_COUNT
```

8.4 #ifdef, #ifndef, #if, #else, #elif, #endif

These instructions are used for conditional compilation with macros.

Code 8-4 #ifdef, endif Example

```
#define   USE_FUNCTION_A
#ifdef    USE_FUNCTION_A
FunctionA:
  ...
#endif
```

Evaluation of the following `#if` and `#elif` expressions is also supported.

- Literal values that can be expressed as signed 32-bit integers. (Operations are undefined for all other values.)
- Minus signs on literal values. (Addition and subtraction are not supported.)
- `defined(`*macro*`)`
- `!defined(`*macro*`)`

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- Equality signs (==, !=)
- Inequality signs (<, <=, >, >=)
- Bitwise AND (&&) and OR (||)
- Any combination of these expressions.

Code 8-5 #if, #endif Example

```
#if (defined(AAA) && (BBB == 1))
  ...
#endif
```

Note: An error is generated when an expression uses an undefined macro.

8.5 #error

This instruction outputs an error.

Code 8-6 #error Example

```
#error    "error message"
```

8.6 #pragma

The `pragma` instruction configures extended information for the assembler.

8.6.1 bind_symbol (symbol_name , start_index [, end_index])

This binds a symbol name to a register. Specify any symbol name in `symbol_name` and specify the starting and ending positions of registers to bind in `start_index` and `end_index`, respectively. You can bind to input registers, floating-point constant registers, integer registers, or Boolean registers.

If you use a `#pragma bind_symbol(symbol_name, start_index)` definition without specifying `end_index`, `start_index` specifies both the starting and ending positions of the registers to bind and thereby specifies a single register.

When an application loads the executable file for the shader assembly code that defines `bind_symbol`, it can use the defined symbol names as arguments to `glGetAttribLocation`, `glBindAttribLocation`, and other functions, and thereby configure the input registers. In the same way, these can also be used as arguments to `glGetUniformLocation` and other functions to configure the floating-point constant registers, integer registers, and Boolean registers.

The `def`, `defb`, and `defi` instructions cannot be used to define a constant value for any registers already specified by `bind_symbol`, either within the same shader assembly code file or within any objects referenced at link time.

You can specify and configure individual components of floating-point constant registers when configuring those registers. To do so, specify the components in the format "`.xyzw`" after the name in `symbol_name`. Specify only consecutive components in xyzw order. (Although you can configure

subsets of components such as "xy", "yzw", and "zw", you cannot configure nonconsecutive subsets such as "xz", "yw", or "xyw.")

With input registers, you cannot set more than one symbol name for a single input register. Also, the same value must be used for start_index and end_index.

Note: The input register settings determine which vertex attributes are used as inputs to the graphics pipeline. In other words, an input register that does not have a symbol bound to it by these settings is not recognized as an input vertex attribute and thus stores undefined values.

When you specify individual components of an input register, the number of components only affects the value of type that is obtained by the glGetActiveAttrib function. If you set size equal to 1, 2, or 3 in the glVertexAttribPointer function, vertex attribute data is loaded respectively into the x, xy, or xyz components of the corresponding register. The default value is loaded into any component into which vertex attribute data is not loaded. By default, the value for y is 0, z is 0, and w is 1.

Note: The hardware does not operate properly when the vertex shaders have not read any input registers. Reading even a single component of any input register during a single cycle of vertex processing is enough to fulfill this requirement. It is acceptable for this input register to have undefined content.

Note: You can use up to 16 input registers. However, when using the vertex buffer to render, you can only load vertex attribute data into up to 12 input registers. Do not use the vertex buffer when using shader assembly code that separately defines bind_symbol for more than 12 input registers.

Code 8-7 bind_symbol Example, in Assembly Code and Application Code

```
// Assembly language source code
#pragma  bind_symbol ( ModelViewMatrix , c0 , c3 )
#pragma  bind_symbol ( Position , v0 )
#pragma  bind_symbol ( LoopCounter0 , i1 , i1 )
#pragma  bind_symbol ( bFirst , b2 , b2 )
#pragma  bind_symbol ( Scalar.x , c4, c4 )   // c4.x is assigned to the Scalar
symbol

// Application source code
glBindAttribLocation ( program , 0 , ”Position” );
glEnableVertexAttribArray ( 0 );

uniform_location = glGetUniformLocation ( program , “ModelViewMatrix” );
GLfloat matrix[4][4];
glUniform4fv ( uniform_location , 4 , matrix );

GLfloat scalar_value;
```


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```
uniform_location = glGetUniformLocation ( program , "Scalar" );
glUniform1f ( uniform_location , scalar_value );

uniform_location = glGetUniformLocation ( program , "bFirst" );
glUniform1i ( uniform_location , GL_TRUE );

GLint loop_setting[3] = { 4 , 0 , 1 } ;          // loop_count−1 , init , step
uniform_location = glGetUniformLocation ( program , "LoopCounter0" );
glUniform3iv ( uniform_location , loop_setting );
```

8.6.2 output_map (data_name , mapped_register)

This configures the attributes of the data output by the vertex shaders. These settings determine which data to output to the fragment pipeline.

You can specify any of the following for `data_name`.

- `position` Vertex coordinates (x y z w)
- `color` Vertex color (R G B A)
- `texture0` Texture coordinate 0 (u v)
- `texture0w` The third component (w) of texture coordinate 0
- `texture1` Texture coordinate 1 (u v)
- `texture2` Texture coordinate 2 (u v)
- `quaternion` Quaternion (x y z w)
- `view` View vector (x y z)
- `generic` General-purpose attribute (freely definable components)

Specify the corresponding output register in `mapped_register`. Because you can specify individual components of the output register, you can pack multiple attributes into a single register. The `generic` attribute is used with geometry shaders.

Note: After a vertex shader has written to all of the output registers specified by these settings, that vertex shader is forced to end its processing and control moves to the next vertex data operation. (The `end` instruction must be called after all the output registers have been written to.) As a result, instructions might not be run if they come after the last attribute data has been written to the output registers.

Code 8-8 Running an Instruction After Writing to the Output Registers

```
mov       o0, r0
mov       o1, r1 // Execution ends here if only o0 and o1 are specified
end             // The end instruction is required
nop             // This instruction might not run
nop             // This instruction might not run
```

Note: Vertex shaders are required to write to the entirety of all of the registers specified by this setting. (All components—x, y, z, and w—of the specified registers must be written. Even if `output_map` has not set all of the components, they must all be written. Dummy values can be written to components that have not been set.)

Code 8-9 Writing to All Specified Registers

```
#pragma  output_map ( position , o0 )
#pragma  output_map ( color , o1 )
#pragma  output_map ( texture0 , o2.xy )
#pragma  output_map ( texture1 , o3.xy )
#pragma  output_map ( texture2 , o4.xy )

...
mov       o0, r0
mov       o1, r1
mov       o2, r2 // Although only 'xy' is specified for o2, o3, and o4,
mov       o3, r3 // some value must be written to the zw components
mov       o4, r4 //
```

Note: After a vertex shader has written to all of the output registers specified by these settings, you must not use any instructions that read or write the various registers. We do not guarantee operations if you attempt to use register read-write instructions after the last write to an output register.

Code 8-10 Running Instructions Illegally After Writing to the Output Registers

```
mov       o0, r0
mov       o1, r1 // Execution ends here if only o0 and o1 were specified
mov       r0, c0 // Instructions that access registers here are prohibited
end             // The end instruction is required
mov       r1, c1 // Instructions that access registers here are prohibited
```

Note: Vertex shaders cannot write to the same output register more than once. Perform only one write operation on all of an output register's components per processing of each single vector. You cannot use multiple writes to write data one component at a time. We do not guarantee results if you use multiple write operations.

Code 8-11 Writing to a Register Multiple Times

```
// Example of valid processing
mov o0.xy, c0
mov o0.zw, c1.w // Each component has only been written once

// Example of invalid processing
```


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```
mov o0.xy, c0
mov o0, c1.w // o0.x and o0.y have been written twice
```

Note: You can use `output_map` to specify attributes other than `generic` in up to 7 output registers. To set 8 or more attributes other than `generic` in the output registers, you must pack multiple attributes into a single output register. When attributes other than `generic` are set in 8 or more output registers, an `INVALID_OPERATION` error is generated when the `glShaderBinary` function loads the code.

Code 8-12 Packing Multiple Attributes Into a Single Output Register

```
#pragma  output_map ( position , o0 )
#pragma  output_map ( color , o1 )
#pragma  output_map ( texture0 , o2.xy )
#pragma  output_map ( texture1 , o2.zw ) // Pack multiple attributes into o2

...
mov       o0, r0
mov       o1, r1
mov       o2.xy, r2.xy
mov       o2.zw, r3.xxxy
```

8.6.3 suppress_linker_warning_level2_level3

This directive suppresses the output of level 2 and level 3 warnings when objects are linked. When a number of shader objects are being linked, the warnings are suppressed only for the main objects for which this `pragma` is defined. Defining this `pragma` for a reference object has no effect. When there is a warning for code in a subroutine inside a reference object, the `pragma` defined for the calling main object determines whether the warning is suppressed.

This `pragma` has been implemented primarily to suppress level 2 warnings, which correspond to the checker feature for uninitialized registers. With the geometry shader, one can imagine use cases where the configured register values are referenced in the next run of the shader, so there is a possibility for heavy use of implementations that reference uninitialized registers. (For example, referencing registers set in the final phase in one run of the shader near the start of the next run.) There is no problem with this kind of shader implementation, so you can use this `pragma` to suppress the output of a large amount of warnings.

Use this in combination with the linker command-line option feature to specify warning levels.

Level 2 warnings correspond to the uninitialized register checker feature, and level 3 warning correspond to the consistency checker feature. Both checks are performed at the level of main objects. In contrast, level 1 warnings include warnings to check the entire shader code to look for code that would cause the GPU to hang. Level 1 warnings are not suppressed because they involve

a different check target and detection method than the other warning levels, and reference objects are handled differently.

Code 8-13 Using the `suppress_linker_warning_level2_level3 pragma` Directive

```
#pragma  suppress_linker_warning_level2_level3
#pragma  suppress_linker_warning_level2_level3 abc      // Error.
// An error occurs if anything unnecessary follows the pragma directive.
```

8.7 #line

This instruction changes the line number and the filename using the following syntax.

`#line` *`line_number`* [“*`filename`*”]

There must be a space between the line number and filename. Use double quotes around the filename and specify a line number that can be represented as a 32-bit integer (`int`) greater than or equal to 1. Operations are undefined for any other value. Use a filename that does not contain any spaces, is composed of no more than 128 single-byte alphanumeric characters, and does not use the following symbols:

`\ / : * ? ” < > |`

Code 8-14 #line Example

```
#line 100 “newname.vsh”
```


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9 Assembly Language Instruction Reference

9.1 Define Instructions

A define instruction sets a constant register value. The value does not change while the shader is running. Regardless of where it is declared in shader assembly code, it is valid throughout the entire linked object. You cannot set more than one value for the same register within the same set of linked objects. You also cannot use define instructions on registers already specified with `#pragma bind_symbol` within the same set of linked objects.

9.1.1 `def` : Define Floating-Point Constants

9.1.1.1 Calling Format

```
def       dest , value0 , value1 , value2 , value3
```

9.1.1.2 Operands

- `dest`: Floating-point constant register
- `value0-4`: Floating-point values

9.1.1.3 Overview

Sets the value of a floating-point constant register. You can set the value in decimal notation (using a decimal point) or integer notation (base 10 or base 16). When you specify values in hexadecimal notation, the hexadecimal bitmap specifies a 24-bit floating-point value (1 sign bit, 7 exponent bits, 16 significand bits). If a value larger than 24 bits is specified, the lower 24 bits are used and the rest is thrown away.

9.1.1.4 Example

```
def       c0 , 1 , 1.5 , -0.5 , 0.25
def       c1 , 0x3f0000 , 0x3f8000 , 0xbe0000 , 0x3d0000  // This sets the same
                                                          // values as above.
```

9.1.2 `defb` : Define Boolean Constant

9.1.2.1 Calling Format

```
defb      dest , value
```

9.1.2.2 Operands

- `dest`: Boolean register
- `value`: `true` or `false`

9.1.2.3 Overview

Sets the value of a Boolean register.

9.1.2.4 Example

```
defb      b0 , true
defb      b1 , false
```

9.1.3 `defi` : Define Integer Constants

9.1.3.1 Calling Format

```
defi      dest , count , init , step
```

9.1.3.2 Operands

- `dest`: Integer register
- `count`: Integer value
- `init`: Integer value
- `step`: Integer value

9.1.3.3 Overview

Sets the value of an integer register. Integer registers are used by the `loop` instruction. The `count` operand specifies a value that is one less than the number of times to run the group of instructions between the `loop` and `endloop` instructions. The `init` operand specifies the initial value of the loop-counter register. The `step` operand specifies the amount by which to increment the loop-counter register during each loop iteration. Specify a value in the range `[0, 255]` for `count`. Specify a value in the range `[-128, 127]` for both `init` and `step`. When its integer value is specified in hexadecimal, `init` and `step` both are specified as two's complement numbers.

Note: For `init`, you can also specify a value in the range `[128, 255]`. However, this is only to maintain compatibility with past versions, so use a value in the range `[-128, 127]`. With the loop counter register set to the `init` value, the hardware implementation is an unsigned 8-bit value. If there is an overflow when you perform a register index offset or add `step`, only those lower 8 bits are valid. The result would be the same, even if the loop counter register was treated as a signed 8-bit value. In other words, specifying the `init` value in the range `[128, 255]` is the same as specifying it in the range `[-128, -1]`.

9.1.3.4 Example

```
defi      i0 , 8 , 0 , 1  // Loops 9 times. aL has an initial value of 0 and is
                          // incremented by one during each loop iteration.
defi      i1 , 10 , 4 , 2 // Loops 11 times. aL has an initial value of 4 and is
                          // incremented by two during each loop iteration.
```

9.2 Arithmetic Instructions

Arithmetic instructions run arithmetic computations. In all these instructions, the components x, y, z, and w in the `src` operands are swizzled before the instruction operation.

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9.2.1 `add` : Add

9.2.1.1 Calling Format

```
add       dest , src0 , src1
```

9.2.1.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.2.1.3 Overview

Stores the sum of `src0` and `src1` in `dest`. You cannot specify a floating-point constant register for both `src0` and `src1`. Nor can you specify input registers with different indices in `src0` and `src1` at the same time.

9.2.1.4 Operation

```
dest.x = src0.x + src1.x
dest.y = src0.y + src1.y
dest.z = src0.z + src1.z
dest.w = src0.w + src1.w
```

9.2.1.5 Example

```
add       r0 , c1 , v2
add       o0.xy , r7.yz , c4.xx
```

9.2.2 `dp3`: Three-Component Dot Product

9.2.2.1 Calling Format

```
dp3       dest , src0 , src1
```

9.2.2.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.2.2.3 Overview

Stores the dot product of three components of `src0` and `src1` in `dest`. You cannot specify a floating-point constant register for both `src0` and `src1`. Nor can you specify input registers with different indices in `src0` and `src1` at the same time.

9.2.2.4 Operation

```
dot = ( src0.x × src1.x ) + ( src0.y × src1.y ) + ( src0.z × src1.z )
dest.x = dot
dest.y = dot
```

```
dest.z = dot
dest.w = dot
```

9.2.2.5 Example

```
dp3       r0 , c1 , v2
dp3       o0 , r7.yzw , c4.xxy
```

9.2.3 dp4: Four-Component Dot Product

9.2.3.1 Calling Format

```
dp4       dest , src0 , src1
```

9.2.3.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.2.3.3 Overview

Stores the dot product of four components of `src0` and `src1` in `dest`. You cannot specify a floating-point constant register for both `src0` and `src1`. Nor can you specify input registers with different indices in `src0` and `src1` at the same time.

9.2.3.4 Operation

```
dot = ( src0.x × src1.x ) + ( src0.y × src1.y ) + ( src0.z × src1.z ) +
( src0.w × src1.w )
dest.x = dot
dest.y = dot
dest.z = dot
dest.w = dot
```

9.2.3.5 Example

```
dp4       r0 , c1 , v2
dp4       o0 , r7.yzwx , c4.xxyw
```

9.2.4 dph: Homogeneous Dot Product

9.2.4.1 Calling Format

```
dph       dest , src0 , src1
```

9.2.4.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

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9.2.4.3 Overview

Stores in `dest` the result of adding the w component of `src1` to the dot product of three components of `src0` and `src1`. You cannot specify a floating-point constant register for both `src0` and `src1`. Nor can you specify input registers with different indices in `src0` and `src1` at the same time.

9.2.4.4 Operation

```
dot = ( src0.x × src1.x ) + ( src0.y × src1.y ) + ( src0.z × src1.z ) +
src1.w
dest.x = dot
dest.y = dot
dest.z = dot
dest.w = dot
```

9.2.4.5 Example

```
dph       r0 , c1 , v2
dph       o0 , r7.yzwx , c4.xxyw
```

9.2.5 `dst`: Distance Vector

9.2.5.1 Calling Format

```
dst       dest , src0 , src1
```

9.2.5.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.2.5.3 Overview

Calculates the vector distance. Sets the y and z components of `src0` to the squared distance, and the y and w components of `src1` to the reciprocal of the distance. The components of `dest` respectively store `1`, the distance, the squared distance, and the reciprocal of the distance. You cannot specify a floating-point constant register for both `src0` and `src1`. Nor can you specify input registers with different indices in `src0` and `src1` at the same time.

9.2.5.4 Operation

```
dest.x = 1
dest.y = src0.y × src1.y
dest.z = src0.z
dest.w = src1.w
```

9.2.5.5 Example

```
dst       r0 , c1 , v2
```

9.2.6 `exp`: Exponential Base 2

9.2.6.1 Calling Format

```
exp       dest , src{ .x | .y | .z | .w }
```

9.2.6.2 Operands

- `dest`: Output register or temporary register
- `src`: Temporary register, input register, or floating-point constant register

9.2.6.3 Overview

Calculates a power of 2. Because this instruction can only calculate a single component, you must specify only one component in `src`.

When this instruction performs calculations that result in ∞ or -∞, sometimes `NaN` are output instead. Use a `cmp` instruction if you need to distinguish between ∞, -∞, and `NaN`.

9.2.6.4 Operation

```
tmp = src { .x | .y | .z | .w }
dest.x = 2 ^ tmp
dest.y = 2 ^ tmp
dest.z = 2 ^ tmp
dest.w = 2 ^ tmp
```

```
exp       r0 , c1.x
```

9.2.7 `flr`: Floor

9.2.7.1 Calling Format

```
flr       dest , src
```

9.2.7.2 Operands

- `dest`: Output register or temporary register
- `src`: Temporary register, input register, or floating-point constant register

9.2.7.3 Overview

Stores the largest integer less than or equal to `src` in `dest`.

9.2.7.4 Operation

```
dest.x = floor ( src.x )
dest.y = floor ( src.y )
dest.z = floor ( src.z )
dest.w = floor ( src.w )
```

9.2.7.5 Example

```
flr       r0, r1
```


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9.2.8 `litp`: Light Coefficients

9.2.8.1 Calling Format

```
litp      dest , src
```

9.2.8.2 Operands

- `dest`: Output register or temporary register
- `src`: Temporary register, input register, or floating-point constant register

9.2.8.3 Overview

Partially calculates lighting. This instruction also changes the status registers at the same time. When performing lighting with vertex shaders, the necessary N·L and N·H sign determination and the clamping of specular item exponents (the value specified for `GL_SHININESS` for material color in OpenGL ES1.1) are performed.

9.2.8.4 Operation

```
dest.x = ( src.x < 0 ) ? 0 : src.x
dest.y = ( src.y < -127.9961 ) ? -127.9961 :
                  ( src.y > 127.9961 ? 127.9961 : src.y )
dest.z = 0
dest.w = ( src.w < 0 ) ? 0 : src.w
status_reg0 = ( src.x > 0 ) ? 1 : 0
status_reg1 = ( src.w > 0 ) ? 1 : 0
```

9.2.8.5 Example

```
// r0.x : Store the value of N·L
// r0.y : Store the specular exponent
// r0.z : This value is not used
// r0.w : Store the value of N·H

litp      r1 , r0
ifc       1 , 1 , 1              // if ( N·H > 0 && N·L > 0 )
pow       r2x , r1.w , r1.y      // r2.x = power ( r1.w , r1.y );
endif
```

9.2.9 `log`: Logarithm Base 2

9.2.9.1 Calling Format

```
log       dest , src{ .x | .y | .z | .w }
```

9.2.9.2 Operands

- `dest`: Output register or temporary register
- `src`: Temporary register, input register, or floating-point constant register

9.2.9.3 Overview

Calculates the base-2 logarithm. Because this instruction can only calculate a single component, you must specify only one component in `src`.

When this instruction performs calculations that result in ∞ or -∞, sometimes `NaN` is output instead. Use a `cmp` instruction if you need to distinguish between ∞, -∞, and `NaN`.

9.2.9.4 Operation

```
tmp = src { .x | .y | .z | .w }
dest.x = log2 ( tmp )
dest.y = log2 ( tmp )
dest.z = log2 ( tmp )
dest.w = log2 ( tmp )
```

9.2.9.5 Example

```
log       r0 , c1.x
```

9.2.10 `mad`: Multiply and Add

9.2.10.1 Calling Format

```
mad       dest , src0 , src1 , src2
```

9.2.10.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register or input register
- `src1`: Temporary register, input register, or floating-point constant register
- `src2`: Temporary register, input register, or floating-point constant register

9.2.10.3 Overview

Stores in `dest` the result of adding `src2` to the product of `src0` and `src1`. You cannot specify a floating-point constant register for both `src1` and `src2`. Nor can you specify input registers with different indices in `src0`, `src1`, and `src2` at the same time.

9.2.10.4 Operation

```
dest.x = src0.x × src1.x + src2.x
dest.y = src0.y × src1.y + src2.y
dest.z = src0.z × src1.z + src2.z
dest.w = src0.w × src1.w + src2.w
```

9.2.10.5 Example

```
mad       r0 , r1, c1 , v2
```


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9.2.11 `max`: Maximum

9.2.11.1 Calling Format

```
max       dest , src0 , src1
```

9.2.11.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.2.11.3 Overview

Compares `src0` and `src1` and stores the larger value in `dest`. You cannot specify a floating-point constant register for both `src0` and `src1`. Nor can you specify input registers with different indices in `src0` and `src1` at the same time.

9.2.11.4 Operation

```
dest.x = ( src0.x > src1.x ) ? src0.x : src1.x
dest.y = ( src0.y > src1.y ) ? src0.y : src1.y
dest.z = ( src0.z > src1.z ) ? src0.z : src1.z
dest.w = ( src0.w > src1.w ) ? src0.w : src1.w
```

9.2.11.5 Example

```
max       r0 , r1 , c1
```

9.2.12 `min`: Minimum

9.2.12.1 Calling Format

```
min       dest , src0 , src1
```

9.2.12.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.2.12.3 Overview

Compares `src0` and `src1` and stores the smaller value in `dest`. You cannot specify a floating-point constant register for both `src0` and `src1`. Nor can you specify input registers with different indices in `src0` and `src1` at the same time.

9.2.12.4 Operation

```
dest.x = ( src0.x > src1.x ) ? src1.x : src0.x
dest.y = ( src0.y > src1.y ) ? src1.y : src0.y
dest.z = ( src0.z > src1.z ) ? src1.z : src0.z
dest.w = ( src0.w > src1.w ) ? src1.w : src0.w
```

9.2.12.5 Example

```
min       r0 , r1 , c1
```

9.2.13 `mov`: Move

9.2.13.1 Calling Format

```
mov       dest , src
```

9.2.13.2 Operands

- `dest`: Output register or temporary register
- `src`: Temporary register, input register, or floating-point constant register

9.2.13.3 Overview

Copies the content of `src` into `dest`.

9.2.13.4 Operation

`dest = src`

9.2.13.5 Example

```
mov       r0 , c1
```

9.2.14 `mova`: Move to Address Register

9.2.14.1 Calling Format

```
mova      dest{ .x | .y | .xy } , src
```

9.2.14.2 Operands

- `dest`: Address register
- `src`: Temporary register, input register, or floating-point constant register

9.2.14.3 Overview

Copies the content of `src` into `dest`. The fractional part of `src` (everything after the decimal point) is truncated. Operations are undefined if you assign a value that does not lie in the range `[-95, 95]`. The address register must use a mask of `.x`, `.y`, or `.xy`. Consecutive calls to this instruction result in an error.

9.2.14.4 Operation

`dest = src`

9.2.14.5 Example

```
mova      a0.x , c1
```


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9.2.15 `mul`: Multiply

9.2.15.1 Calling Format

```
mul       dest , src0 , src1
```

9.2.15.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.2.15.3 Overview

Stores the product of `src0` and `src1` in `dest`. You cannot specify a floating-point constant register for both `src0` and `src1`. Nor can you specify input registers with different indices in `src0` and `src1` at the same time.

9.2.15.4 Operation

```
dest.x = src0.x × src1.x
dest.y = src0.y × src1.y
dest.z = src0.z × src1.z
dest.w = src0.w × src1.w
```

9.2.15.5 Example

```
mul       r0 , c1 , v2
mul       o0.xy , r7.yzww , c4.xxyz
```

9.2.16 `nop`: No Operation

9.2.16.1 Calling Format

```
nop
```

9.2.16.2 Operands

None

9.2.16.3 Overview

This instruction does nothing.

9.2.16.4 Operation

None

9.2.16.5 Example

```
nop
```

9.2.17 `rcp`: Reciprocal

9.2.17.1 Calling Format

```
rcp       dest , src{ .x | .y | .z | .w }
```

9.2.17.2 Operands

- `dest`: Output register or temporary register
- `src`: Temporary register, input register, or floating-point constant register

9.2.17.3 Overview

Calculates the reciprocal. Because this instruction can only calculate a single component, you must specify only one component in `src`.

When this instruction performs calculations that result in ∞ or -∞, sometimes `NaN` is output instead. Use a `cmp` instruction if you need to distinguish between ∞, -∞, and `NaN`.

9.2.17.4 Operation

```
tmp = src { .x | .y | .z | .w }
dest.x = 1 / tmp
dest.y = 1 / tmp
dest.z = 1 / tmp
dest.w = 1 / tmp
```

9.2.17.5 Example

```
rcp       r0 , c1.x
```

9.2.18 `rsq`: Reciprocal Square Root

9.2.18.1 Calling Format

```
rsq       dest , src{ .x | .y | .z | .w }
```

9.2.18.2 Operands

- `dest`: Output register or temporary register
- `src`: Temporary register, input register, or floating-point constant register

9.2.18.3 Overview

Calculates the square root of the reciprocal. Because this instruction can only calculate a single component, you must specify only one component in `src`.

When this instruction performs calculations that result in ∞ or -∞, sometimes `NaN` is output instead. Use a `cmp` instruction if you need to distinguish between ∞, -∞, and `NaN`.

9.2.18.4 Operation

```
tmp = src { .x | .y | .z | .w }
dest.x = 1 / sqrt ( tmp )
dest.y = 1 / sqrt ( tmp )
```


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```
dest.z = 1 / sqrt ( tmp )
dest.w = 1 / sqrt ( tmp )
```

9.2.18.5 Example

```
rsq       r0 , c1.x
```

9.2.19 `sge`: Set on Greater Than or Equal

9.2.19.1 Calling Format

```
sge       dest , src0 , src1
```

9.2.19.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.2.19.3 Overview

Stores `1` in `dest` when `src0` is greater than or equal to `src1`; otherwise, stores `0` in `dest`. You cannot specify a floating-point constant register for both `src0` and `src1`. Nor can you specify input registers with different indices in `src0` and `src1` at the same time.

9.2.19.4 Operation

```
dest.x = ( src0.x >= src1.x ) ? 1 : 0
dest.y = ( src0.y >= src1.y ) ? 1 : 0
dest.z = ( src0.z >= src1.z ) ? 1 : 0
dest.w = ( src0.w >= src1.w ) ? 1 : 0
```

9.2.19.5 Example

```
sge       r0 , c1 , v2
sge       o0.xy , r7.yzww , c4.xxyz
```

9.2.20 `slt`: Set on Less Than

9.2.20.1 Calling Format

```
slt       dest , src0 , src1
```

9.2.20.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.2.20.3 Overview

Stores `1` in `dest` when `src0` is smaller than `src1`; otherwise, stores `0` in `dest`. You cannot specify a floating-point constant register for both `src0` and `src1`. Nor can you specify input registers with different indices in `src0` and `src1` at the same time.

9.2.20.4 Operation

```
dest.x = ( src0.x < src1.x ) ? 1 : 0
dest.y = ( src0.y < src1.y ) ? 1 : 0
dest.z = ( src0.z < src1.z ) ? 1 : 0
dest.w = ( src0.w < src1.w ) ? 1 : 0
```

9.2.20.5 Example

```
slt       r0 , c1 , v2
slt       o0.xy , r7.yzww , c4.xxyz
```

9.3 Macro Instructions

Macro instructions expand into a combination of arithmetic instructions.

9.3.1 `sub`: Subtract

9.3.1.1 Calling Format

```
sub       dest , src0 , src1
```

9.3.1.2 Operands

- `dest`: Output register or temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.3.1.3 Overview

Stores the difference of `src0` and `src1` in dest. You cannot specify a floating-point constant register for both `src0` and `src1`. Nor can you specify input registers with different indices in `src0` and `src1` at the same time.

9.3.1.4 Operation

```
dest.x = src0.x - src1.x
dest.y = src0.y - src1.y
dest.z = src0.z - src1.z
dest.w = src0.w - src1.w
```

9.3.1.5 Post-Macro Expansion

```
add       dest , src0 , -src1
```

9.3.1.6 Example

```
sub       r0 , c1 , v2
```


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```
sub       o0.xy , r7.yzww , c4.xxyz
```

9.3.2 abs : Absolute

9.3.2.1 Calling Format

```
abs       dest , src
```

9.3.2.2 Operands

- dest: Output register or temporary register
- src: Temporary register or input register

9.3.2.3 Overview

Stores the absolute value of src in dest.

9.3.2.4 Operation

```
dest.x = abs ( src0.x )
dest.y = abs ( src0.y )
dest.z = abs ( src0.z )
dest.w = abs ( src0.w )
```

9.3.2.5 Post-Macro Expansion

```
max       dest , src , -src
```

9.3.2.6 Example

```
abs       r0, r1
abs       o0 , r7.yzww
```

9.3.3 crs: Cross Product

9.3.3.1 Calling Format

```
crs       dest{ .x | .y | .z | .xy | .xz | .yz | .xyz } , src0 , src1
```

9.3.3.2 Operands

- dest: Temporary register
- src0: Temporary register, input register, or floating-point constant register
- src1: Temporary register, input register, or floating-point constant register

9.3.3.3 Overview

Stores the cross product of three components of src0 and src1 in dest. Neither src0 nor src1 can be swizzled. You must use one of the following masks for the dest operand: .x, .y, .z, .xy, .xz, .yz, or .xyz.

You cannot specify any of the following.

- The same register for src0 and dest.

- The same register for `src1` and `dest`.
- A mask other than the default (`.xyzw`) for either `src0` or `src1`.
- A floating-point constant register for both `src0` and `src1`.
- Input registers with different indices for `src0` and `src1` at the same time.

9.3.3.4 Operation

```
dest.x = src0.y × src1.z - src0.z × src1.y
dest.y = src0.z × src1.x - src0.x × src1.z
dest.z = src0.x × src1.y - src0.y × src1.x
```

9.3.3.5 Post-Macro Expansion

```
mul       dest.xyz , src0.yzx , src1.zxy
mad       dest.xyz , -src1.yzx , src0.zxy , dest
```

9.3.3.6 Example

```
crs       r0.xyz , c1 , v2
```

9.3.4 `frc`: Fraction

9.3.4.1 Calling Format

```
frc       dest , src
```

9.3.4.2 Operands

- `dest`: Temporary register
- `src`: Temporary register, input register, or floating-point constant register

9.3.4.3 Overview

Stores in `dest` the difference between the value of `src` and the largest integer less than or equal to `src`. You cannot specify the same register for `src` and `dest`.

9.3.4.4 Operation

```
dest.x = src.x - floor ( src.x )
dest.y = src.y - floor ( src.y )
dest.z = src.z - floor ( src.z )
dest.w = src.w - floor ( src.w )
```

9.3.4.5 Post-Macro Expansion

```
flr       dest , src
add       dest , src , −dest
```

9.3.4.6 Example

```
frc       r0 , v1
```


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9.3.5 `lrp`: Linear Interpolation

9.3.5.1 Calling Format

```
lrp       dest , src0 , src1 , src2
```

9.3.5.2 Operands

- `dest`: Temporary register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register
- `src2`: Temporary register, input register, or floating-point constant register

9.3.5.3 Overview

Stores in `dest` the result of using `src0` to linearly interpolate between `src1` and `src2`. You can specify only one floating-point constant register among the three operands `src0`, `src1`, and `src2`. However, there is one exception: you can specify two floating-point constant registers at the same time if you specify them for `src0` and `src1`.

You cannot do any of the following.

- Specify the same register for `dest` and `src0`.
- Use the same register for `dest` and `src2`.
- Specify input registers with different indices in any combination of `src0`, `src1`, and `src2` at the same time.

9.3.5.4 Operation

```
dest.x = src0.x × src1.x + (1 − src0.x )×src2.x
dest.y = src0.y × src1.y + (1 − src0.y )×src2.y
dest.z = src0.z × src1.z + (1 − src0.z )×src2.z
dest.w = src0.w × src1.w + (1 − src0.w )×src2.w
```

9.3.5.5 Post-Macro Expansion

```
add       dest , src1 , −src2
mad       dest , dest , src0 , src2
```

9.3.5.6 Example

```
lrp       r0 , v1 , c2 , r3
```

9.3.6 `m3x2`: 3x2 Multiply

9.3.6.1 Calling Format

```
m3x2      dest.xy , src0 , src1
```

9.3.6.2 Operands

- `dest`: Temporary register or output register

- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.3.6.3 Overview

Stores the result of multiplying a 3x2 matrix and a 3-component vector in `dest`. Specify the first register of the 3x2 matrix in `src1` (in other words, when `src1` is `r0`, that means the 3x2 matrix is stored in `r0` and `r1`).

You cannot specify any of the following.

- A floating-point constant register for both `src0` and `src1`.
- An input register for both `src0` and `src1`.
- The same register for `dest` and `src0`.

You must use the mask `.xy` for `dest`.

Note: If you set `dest` to the register that consecutively follows `src1` (`src2` in the expanded macro below), the content of that register is updated after macro expansion when the instruction executes, causing unexpected results.

9.3.6.4 Operation

```
src2 = Next_Index_Of ( src1 )
dest.x = ( src0.x × src1.x ) + ( src0.y × src1.y ) + ( src0.z × src1.z )
dest.y = ( src0.x × src2.x ) + ( src0.y × src2.y ) + ( src0.z × src2.z )
```

9.3.6.5 Post-Macro Expansion

```
dp3       dest.x , src0 , src1
dp3       dest.y , src0 , src2           // src2 = next index after src1
```

9.3.6.6 Example

```
m3x2      r0.xy , r1 , c0
// This is expanded as follows
// dp3    r0.x , r1 , c0
// dp3    r0.y , r1 , c1
```

9.3.7 `m3x3`: 3x3 Multiply

9.3.7.1 Calling Format

```
m3x3      dest.xyz , src0 , src1
```

9.3.7.2 Operands

- `dest`: Temporary register or output register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

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9.3.7.3 Overview

Stores the result of multiplying a 3x3 matrix and a 3-component vector in `dest`. Specify the first register of the 3x3 matrix in `src1` (in other words, when `src1` is `r0`, that means the 3x3 matrix is stored in `r0`, `r1`, and `r2`).

You cannot specify any of the following.

- A floating-point constant register for both `src0` and `src1`.
- An input register for both `src0` and `src1`.
- The same register for `dest` and `src0`.

You must use the mask `.xyz` for `dest`.

Note: If you set `dest` to a register that consecutively follows `src1` (`src2` and `src3` in the expanded macro below), the content of that register is updated after macro expansion when the instruction executes, causing unexpected results.

9.3.7.4 Operation

```
src2 = Next_Index_Of ( src1 )
src3 = Next_Index_Of ( src2 )
dest.x = ( src0.x × src1.x ) + ( src0.y × src1.y ) + ( src0.z × src1.z )
dest.y = ( src0.x × src2.x ) + ( src0.y × src2.y ) + ( src0.z × src2.z )
dest.z = ( src0.x × src3.x ) + ( src0.y × src3.y ) + ( src0.z × src3.z )
```

9.3.7.5 Post-Macro Expansion

```
dp3       dest.x , src0 , src1
dp3       dest.y , src0 , src2          // src2 = next index after src1
dp3       dest.z , src0 , src3          // src3 = next index after src2
```

9.3.7.6 Example

```
m3x3      r0.xyz , r1 , c0
// This is expanded as follows
// dp3    r0.x , r1 , c0
// dp3    r0.y , r1 , c1
// dp3    r0.z , r1 , c2
```

9.3.8 `m3x4`: 3x4 Multiply

9.3.8.1 Calling Format

```
m3x4      dest , src0 , src1
```

9.3.8.2 Operands

- `dest`: Temporary register or output register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.3.8.3 Overview

Stores the result of multiplying a 3x4 matrix and a 3-component vector in `dest`. Specify the first register of the 3x4 matrix in `src1` (in other words, when `src1` is `r0`, that means the 3x4 matrix is stored in `r0`, `r1`, `r2`, and `r3`).

You cannot specify any of the following.

- A floating-point constant register for both `src0` and `src1`.
- An input register for both `src0` and `src1`.
- A mask other than the default (`.xyzw`) for `dest`.
- The same register for `dest` and `src0`.

Note: If you set `dest` to a register that consecutively follows `src1` (`src2, src3,` and `src4` in the expanded macro below), the content of that register is updated after macro expansion when the instruction executes, causing unexpected results.

9.3.8.4 Operation

```
src2 = Next_Index_Of ( src1 )
src3 = Next_Index_Of ( src2 )
src4 = Next_Index_Of ( src3 )
dest.x = ( src0.x × src1.x ) + ( src0.y × src1.y ) + ( src0.z × src1.z )
dest.y = ( src0.x × src2.x ) + ( src0.y × src2.y ) + ( src0.z × src2.z )
dest.z = ( src0.x × src3.x ) + ( src0.y × src3.y ) + ( src0.z × src3.z )
dest.w = ( src0.x × src4.x ) + ( src0.y × src4.y ) + ( src0.z × src4.z )
```

9.3.8.5 Post-Macro Expansion

```
dp3       dest.x , src0 , src1
dp3       dest.y , src0 , src2          // src2 = next index after src1
dp3       dest.z , src0 , src3          // src3 = next index after src2
dp3       dest.w , src0 , src4          // src4 = next index after src3
```

9.3.8.6 Example

```
m3x4      r0 , r1 , c0
// This is expanded as follows
// dp3    r0.x , r1 , c0
// dp3    r0.y , r1 , c1
// dp3    r0.z , r1 , c2
// dp3    r0.w , r1 , c3
```

9.3.9 `m4x3`: 4x3 Multiply

9.3.9.1 Calling Format

```
m4x3      dest.xyz , src0 , src1
```


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9.3.9.2 Operands

- `dest`: Temporary register or output register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.3.9.3 Overview

Stores the result of multiplying a 4x3 matrix and a 4-component vector in `dest`. Specify the first register of the 4x3 matrix in `src1` (in other words, when `src1` is `r0`, that means the 4x3 matrix is stored in `r0`, `r1`, and `r2`).

You cannot specify any of the following.

- A floating-point constant register for both `src0` and `src1`.
- An input register for both `src0` and `src1`.
- The same register for `dest` and `src0`.

You must use the mask `.xyz` for `dest`.

Note: If you set `dest` equal to a register that consecutively follows `src1` (`src2` and `src3` in the expanded macro below), the content of that register is updated after macro expansion when the instruction executes, causing unexpected results.

9.3.9.4 Operation

```
src2 = Next_Index_Of ( src1 )
src3 = Next_Index_Of ( src2 )
dest.x = ( src0.x × src1.x ) + ( src0.y × src1.y ) + ( src0.z × src1.z ) +
( src0.w × src1.w )
dest.y = ( src0.x × src2.x ) + ( src0.y × src2.y ) + ( src0.z × src2.z ) +
( src0.w × src2.w )
dest.z = ( src0.x × src3.x ) + ( src0.y × src3.y ) + ( src0.z × src3.z ) +
( src0.w × src3.w )
```

9.3.9.5 Post-Macro Expansion

```
dp4       dest.x , src0 , src1
dp4       dest.y , src0 , src2          // src2 = next index after src1
dp4       dest.z , src0 , src3          // src3 = next index after src2
```

9.3.9.6 Example

```
m4x3      r0.xyz , r1 , c0
// This is expanded as follows
// dp4    r0.x , r1 , c0
// dp4    r0.y , r1 , c1
// dp4    r0.z , r1 , c2
```

9.3.10 `m4x4`: 4x4 Multiply

9.3.10.1 Calling Format

```
m4x4      dest , src0 , src1
```

9.3.10.2 Operands

- `dest`: Temporary register or output register
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.3.10.3 Overview

Stores the result of multiplying a 4x4 matrix and a 4-component vector in `dest`. Specify the first register of the 4x4 matrix in `src1` (in other words, when `src1` is `r0`, that means the 4x4 matrix is stored in `r0`, `r1`, `r2`, and `r3`).

You cannot specify any of the following.

- A floating-point constant register for both `src0` and `src1`.
- An input register for both `src0` and `src1`.
- The same register for `dest` and `src0`.

You cannot use a mask other than the default (`.xyzw`) for `dest`.

Note: If you set `dest` equal to a register that consecutively follows `src1` (`src2`, `src3`, and `src4` in the expanded macro below), the content of that register is updated after macro expansion when the instruction executes, causing unexpected results.

9.3.10.4 Operation

```
src2 = Next_Index_Of ( src1 )
src3 = Next_Index_Of ( src2 )
src4 = Next_Index_Of ( src3 )
dest.x = ( src0.x × src1.x ) + ( src0.y × src1.y ) + ( src0.z × src1.z ) +
( src0.w × src1.w )
dest.y = ( src0.x × src2.x ) + ( src0.y × src2.y ) + ( src0.z × src2.z ) +
( src0.w × src2.w )
dest.z = ( src0.x × src3.x ) + ( src0.y × src3.y ) + ( src0.z × src3.z ) +
( src0.w × src3.w )
dest.w = ( src0.x × src4.x ) + ( src0.y × src4.y ) + ( src0.z × src4.z ) +
( src0.w × src4.w )
```

9.3.10.5 Post-Macro Expansion

```
dp4       dest.x , src0 , src1
dp4       dest.y , src0 , src2          // src2 = next index after src1
dp4       dest.z , src0 , src3          // src3 = next index after src2
dp4       dest.w , src0 , src4          // src4 = next index after src3
```


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9.3.10.6 Example

```
m4x3      r0 , r1 , c0
// This is expanded as follows
// dp4    r0.x , r1 , c0
// dp4    r0.y , r1 , c1
// dp4    r0.z , r1 , c2
// dp4    r0.w , r1 , c3
```

9.3.11 nrm: Normalize

9.3.11.1 Calling Format

```
nrm       dest , src
```

9.3.11.2 Operands

- `dest`: Temporary register
- `src`: Temporary register or input register

9.3.11.3 Overview

Stores the result of normalizing `src` in `dest`. You cannot specify the same register for `src` and `dest`.

9.3.11.4 Operation

```
tmp = sqrt ( src.x × src.x + src.y × src.y + src.z × src.z + src.w ×
src.w )
dest.x = src.x× ( 1 / tmp )
dest.y = src.y× ( 1 / tmp )
dest.z = src.z× ( 1 / tmp )
dest.w = src.w× ( 1 / tmp )
```

9.3.11.5 Post-Macro Expansion

```
dp4       dest.x , src , src
rsq       dest.x , dest.x
mul       dest , src , dest.x
```

9.3.11.6 Example

```
nrm       r0 , v0
```

9.3.12 pow : Power

9.3.12.1 Calling Format

```
pow       dest , src0{ .x | .y | .z | .w} , src1{ .x | .y | .z | .w}
```

9.3.12.2 Operands

- `dest`: Temporary register

- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.3.12.3 Overview

Stores the result of raising `src0` to the `src1` power in `dest`. You must specify one of the four components (`.x`, `.y`, `.z`, or `.w`) for `src0` and `src1`. You cannot specify the same register for `src1` and `dest`.

9.3.12.4 Operation

```
tmp = src0{ .x | .y | .z | .w} ^ src1{ .x | .y | .z | .w}
dest.x = tmp
dest.y = tmp
dest.z = tmp
dest.w = tmp
```

9.3.12.5 Post-Macro Expansion

```
log       dest.z , src0{ .x | .y | .z | .w}
mul       dest.z , dest.z , src1{ .x | .y | .z | .w}
exp       dest , dest.z
```

9.3.12.6 Example

```
pow       r0 , r1.y , r2.x
```

9.3.13 `sgn`: Sign

9.3.13.1 Calling Format

```
sgn       dest , src0 , src1 , src2
```

9.3.13.2 Operands

- `dest`: Temporary register or output register
- `src0`: Temporary register or input register
- `src1`: Temporary register
- `src2`: Temporary register

9.3.13.3 Overview

Stores 1, 0, or -1 in `dest` when `src0` is positive, zero, or negative, respectively. `src1` and `src2` are used to perform the calculations. You cannot use swizzling or specify negative signs with `src1` and `src2`. You cannot specify the same register for `src1` and `src2`. You cannot specify the same register for `src0` and `src1`.

9.3.13.4 Operation

```
for ( each component )
{
  if ( src0.component < 0 )
```


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```
            dest.component = -1
    else if ( src0.component == 0 )
            dest.component = 0
    else
            dest.component = 1
}
```

9.3.13.5 Post-Macro Expansion

```
slt       src1 , src0 , -src0
slt       src2 , -src0 , src0
add       dest , src2 , -src1
```

9.3.13.6 Example

```
sgn       r0 , v1 , r2 , r3
```

9.3.14 `sincos`: Sine and Cosine

9.3.14.1 Calling Format

```
sincos            dest{ .x | .y | .xy } , src0{ .x | .y | .z | .w } , src1 , src2
```

9.3.14.2 Operands

- `dest`: Temporary register
- `src0`: Temporary register or input register
- `src1`: Temporary register
- `src2`: Temporary register

9.3.14.3 Overview

Calculates the sine and cosine of the component value specified by `src0` and stores the result in `dest`. Values are specified in radians. The cosine and sine values are output to the x and y components, respectively, in `dest`. You must specify one of the following masks for `dest`: `.x`, `.y`, or `.xy`. The content of the z component in `dest` is not preserved because it is used during calculations. You must specify one of the four components (`.x`, `.y`, `.z`, or `.w`) for `src0`. `src1` and `src2` are used to perform the calculations. You cannot use swizzling or specify negative signs with `src1` and `src2`. A different register must be specified for each operand. The component specified by `src0` must have a value between $-\pi$ and π. This macro calculates an approximate value using a Taylor expansion. Because a Taylor expansion requires several coefficients, the floating-point constant registers `c93`, `c94`, and `c95` are automatically defined. You cannot use this instruction and define `c93`, `c94`, or `c95` with the `def` instruction.

9.3.14.4 Operation

```
tmp = src0{ .x | .y | .z | .w }
dest.x = cos ( tmp )
```

```
dest.y = sin ( tmp )
dest.z = undefined     // This is used during calculations
```

9.3.14.5 Post-Macro Expansion

```
def   c93 , 0xbe0000, 0xbc5555, 0x3f0000, 0x3f0000   // -0.5, -1/3!, 1.0, 1.0
def   c94 , 0xb56c16, 0xb2a01a, 0x3a5555, 0x381111   // -1/6!, -1/7!, 1/4!, 1/5!
def   c95 , 0xa927e4, 0xa5ae64, 0x2fa01a, 0x2c71de   // -1/10!, -1/11!, 1/8!, 1/9!

mov   src1 , c95
mov   src2 , c94
mul   dest.z , src0{ .x | .y | .z | .w } , src0{ .x | .y | .z | .w }
mad   dest.xy , dest.z , src1.xy , src1.zw
mad   dest.xy , dest.z , dest.xy , src2.xy
mad   dest.xy , dest.z , dest.xy , src2.zw
mov   src1 , c93
mad   dest.xy , dest.z , dest.xy , src1.xy
mad   dest.xy , dest.z , dest.xy , src1.zw
mul   dest.y , dest.y , src0{ .x | .y | .z | .w }
```

9.3.14.6 Example

```
sincos    r0.xy , v1.x , r2 , r3
```

9.4 Flow Control Instructions

The flow control instructions control the flow of execution.

9.4.1 `call`: Call Subroutine

9.4.1.1 Calling Format

```
call      label
```

9.4.1.2 Operands

- `label`: Label name

9.4.1.3 Overview

Causes control to jump to the address of the specified label name. Processing returns to the address immediately following this instruction when the subroutine processing has finished (that is, when the `ret` instruction is encountered after the label address). You cannot call a label if a `ret` instruction has not been set for it. You can nest up to four call instructions (`call`, `callc`, and `callb`). Operations are undefined for five or more nested calls. When call instructions are nested, operations are undefined if a call instruction is invoked immediately before a `ret` instruction.

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9.4.1.4 Operation

```
retaddr = pc + 1
pc = get_label_address ( label )
while (1)
{
  execute_current_instruction ( )
  if ( current_instruction () == ret )
  {
         pc = retaddr
         break
  }
}
```

9.4.1.5 Example

```
call      subfunction0

subfunction0:
..
ret
```

9.4.2 `callb`: Boolean Call

9.4.2.1 Calling Format

```
callb     src, label
```

9.4.2.2 Operands

- `src`: Boolean register
- `label`: Label name

9.4.2.3 Overview

Causes control to jump to the address of the specified label name when the content of the specified Boolean register is `true`. If processing ends before reaching a `ret` instruction after the label address, it returns to the address immediately after this instruction. You cannot call a label if a `ret` instruction has not been set for it. You can nest up to four call instructions (`call`, `callc`, and `callb`). Operations are undefined for five or more nested calls. When call instructions are nested, operations are undefined if a call instruction is invoked immediately before a `ret` instruction.

9.4.2.4 Operation

```
if ( src )
  call   label
```

9.4.2.5 Example

```
callb     b0 , subfunction0
```

```
subfunction0:
..
ret
```

9.4.3 `callc`: Condition Call

9.4.3.1 Calling Format

```
callc     status0 , status1 , mode , label
```

9.4.3.2 Operands

- `status0`: Value (either 0 or 1) of status register 0
- `status1`: Value (either 0 or 1) of status register 1
- `mode`: Conditional mode
  - 0: OR
  - 1: AND
  - 2: Only status register 0
  - 3: Only status register 1
- `label`: Label name

9.4.3.3 Overview

Calls a function conditionally based on the status register values.

The equality of the values specified by `status0` (or `status1`) and status register 0 (or 1) is taken to be the condition. This condition is `true` when either status register 0 or 1 match when `mode` is 0; when both status registers match when `mode` is 1; when status register 0 matches when `mode` is 2; and when status register 1 matches when `mode` is 3.

This instruction causes control to jump to the address of the specified label when the condition is `true`. If processing ends before reaching a `ret` instruction after the label address, it returns to the address immediately after this instruction. You cannot call a label if a `ret` instruction has not been set for it. You can nest up to four call instructions (`call`, `callc`, and `callb`). Operations are undefined for five or more nested calls. When call instructions are nested, operations are undefined if a call instruction is invoked immediately before a `ret` instruction.

9.4.3.4 Operation

```
switch ( mode )
{
case 0 :
  if ( status0 == Status_register0 || status1 == Status_register1 )
         call  label
  break;
case 1 :
  if ( status0 == Status_register0 && status1 == Status_register1 )
```


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```
        call  label
  break;
case 2 :
  if ( status0 == Status_register0 )
        call  label
  break;
case 3 :
  if ( status1 == Status_register1 )
        call  label
  break;
}
```

9.4.3.5 Example

```
Callc   1 , 1 , 0 , subfunction0   // Calls subfunction0 when status register 0
                                   // or status register 1 is equal to 1

subfunction0:
..
ret
```

9.4.4 `jpb`: Boolean Jump

9.4.4.1 Calling Format

```
jpb       src, value , label
```

9.4.4.2 Operands

- `src`: Boolean register
- `value`: `true` or `false`
- `label`: Label name

9.4.4.3 Overview

Causes control to jump to the address of the specified label name when the content of the Boolean register specified by `src` matches the value specified by `value`. Unlike the `call` instruction, control does not return at a `ret` instruction and you can also specify labels without setting a `ret` instruction. Jumping to an external location from within an `if` or `loop` block, or jumping from an external location into within an `if` or `loop` block causes an error. Calling this instruction immediately prior to an `else`, `endif`, `endloop`, or `ret` instruction also causes an error. Jumping to an external location from between the `main` and `endmain` labels and from within subroutines results in undefined operations. In the same way, jumping to a `ret` instruction within a subroutine also results in undefined operations. Jumping to an `else`, `endif`, or `endloop` instruction has the same effect as jumping to the instruction immediately following that instruction.

9.4.4.4 Operation

```
if ( src == value )
  jump to       label
```

9.4.4.5 Example

```
jpb       b0 , true , subfunction0
jpb       b1 , false , subfunction0

subfunction0:
..
```

9.4.5 `jpc`: Condition Jump

9.4.5.1 Calling Format

```
jpc       status0 , status1 , mode , label
```

9.4.5.2 Operands

- `status0`: Value (either 0 or 1) of status register 0
- `status1`: Value (either 0 or 1) of status register 1
- `mode`: Conditional mode
  - 0: OR
  - 1: AND
  - 2: Only status register 0
  - 3: Only status register 1
- `label`: Label name

9.4.5.3 Overview

Causes control to jump conditionally based on the status register values.

The equality of the values specified by `status0` (or `status1`) and status register 0 (or 1) is taken to be the condition. This condition is `true` when either status register 0 or 1 match when `mode` is 0; when both status registers match when `mode` is 1; when status register 0 matches when `mode` is 2; and when status register 1 matches when `mode` is 3.

This instruction causes control to jump to the address of the specified label when the condition is `true`. Unlike the `call` instruction, control does not return at a `ret` instruction and you can also specify labels without setting a `ret` instruction. Jumping to an external location from within an `if` or `loop` block, or jumping from an external location into within an `if` or `loop` block causes an error. Calling this instruction immediately prior to an `else`, `endif`, `endloop`, or `ret` instruction also causes an error. Jumping to an external location from between the `main` and `endmain` labels and from within subroutines results in undefined operations. In the same way, jumping to a `ret` instruction within a subroutine also results in undefined operations. Jumping to an `else`, `endif`, or `endloop`, instruction has the same effect as jumping to the instruction immediately following that instruction.

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9.4.5.4 Operation

```
switch ( mode )
{
case 0 :
  if ( status0 == Status_register0 || status1 == Status_register1 )
         jump to   label
  break;
case 1 :
  if ( status0 == Status_register0 && status1 == Status_register1 )
         jump to   label
  break;
case 2 :
  if ( status0 == Status_register0 )
         jump to   label
  break;
case 3 :
  if ( status1 == Status_register1 )
         jump to   label
  break;
}
```

9.4.5.5 Example

```
jpc   1 , 1 , 0 , subfunction0   // Calls subfunction0 when status register 0 or
                                 // status register 1 is equal to 1

subfunction0:
..
```

9.4.6 `ret`: Return from Subroutine

9.4.6.1 Calling Format

```
ret
```

9.4.6.2 Operands

None

9.4.6.3 Overview

Jumps to the caller of a call instruction. This instruction does nothing if control is not in the middle of a jump from a call instruction (however, unlike `nop`, no processing occurs). To call a label as a subroutine from a call instruction, this instruction must be called after the label is set.

9.4.6.4 Operation

```
if ( retaddr )
  pc = retaddr
```

9.4.6.5 Example

```
callb     b0 , subfunction0

subfunction0:
..
ret
```

9.4.7 `ifb`: Start if Block by Boolean

9.4.7.1 Calling Format

```
ifb       src
```

9.4.7.2 Operands

- `src`: Boolean register

9.4.7.3 Overview

Executes conditional processing based on the content of the Boolean register specified by `src`. When it is `true`, the instructions between `ifb` and `endif` are executed. If there is an `else` instruction between `ifb` and `endif`, the instructions between `ifb` and `else` are executed. When it is `false`, the instructions between `ifb` and `endif` are skipped and control moves to the instruction immediately after `endif`. If there is an `else` instruction between `ifb` and `endif`, the instructions between `else` and `endif` are executed. This instruction must be followed by an `endif` instruction. You can nest up to eight `ifb` and `ifc` instructions. You must denote at least one instruction between `ifb` and `endif` as well as between `ifb` and `else`.

9.4.7.4 Operation

```
if ( src == true )
{
  ...
}
```

9.4.7.5 Example

```
ifb       b0
...
endif
```

9.4.8 `ifc`: Start if Block by Condition

9.4.8.1 Calling Format

```
ifc       status0 , status1 , mode
```


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9.4.8.2 Operands

- `status0`: Value (either 0 or 1) of status register 0
- `status1`: Value (either 0 or 1) of status register 1
- `mode`: Conditional mode
  - 0: OR
  - 1: AND
  - 2: Only status register 0
  - 3: Only status register 1

9.4.8.3 Overview

Runs conditional processing based on the status register values.

The equality of the values specified by `status0` (or `status1`) and status register 0 (or 1) is taken to be the condition. This condition is `true` when either status register 0 or 1 match when `mode` is 0; when both status registers match when `mode` is 1; when status register 0 matches when `mode` is 2; and when status register 1 matches when `mode` is 3.

When the condition is `true`, the instructions between `ifc` and `endif` are executed. If there is an `else` instruction between `ifc` and `endif`, the instructions between `ifc` and `else` are executed. When the condition is `false`, the instructions between `ifc` and `endif` are skipped and control moves to the instruction immediately after `endif`. If there is an `else` instruction between `ifc` and `endif`, the instructions between `else` and `endif` are executed. This instruction must be followed by an `endif` instruction. You can nest up to eight `ifb` and `ifc` instructions.You must denote at least one instruction between `ifc` and `endif` as well as between `ifc` and `else`.

9.4.8.4 Operation

```
switch ( mode )
{
case 0 :
  if ( status0 == Status_register0 || status1 == Status_register1 )
         condition = true
  break;
case 1 :
  if ( status0 == Status_register0 && status1 == Status_register1 )
         condition = true
  break;
case 2 :
  if ( status0 == Status_register0 )
         condition = true
  break;
case 3 :
  if ( status1 == Status_register1 )
         condition = true
```

```
  break;
}
if ( condition == true )
{
  ...
}
```

9.4.8.5 Example

```
ifc       1 , 1 , 0       // Runs the instructions between ifc and else when
...                     // status register 0 or status register 1 is equal to 1
else
  ...
endif
..
```

9.4.9 `else`: Start else Block

9.4.9.1 Calling Format

```
else
```

9.4.9.2 Operands

None

9.4.9.3 Overview

This is used in combination with `ifc` or `ifb`. When the `if` statement is `true`, processing runs until this instruction and then skips all instructions until the next `endif`. When the `if` statement is `false`, processing skips from the `if` instruction to this one and then runs all instructions between this and `endif`. You must denote at least one instruction between `ifb` and `else`, between `ifc` and `else`, and between `else` and `endif`.

9.4.9.4 Operation

```
if ( src == true )
{
  ...
}
else
{
  ...
}
endif
```

9.4.9.5 Example

```
ifb       b0
...
```


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```
else
...
endif
```

9.4.10 `endif`: End if Block

9.4.10.1 Calling Format

```
endif
```

9.4.10.2 Operands

None.

9.4.10.3 Overview

Ends a control block started by `ifc` or `ifb`. You must denote at least one instruction between `ifb` and `endif`, between `ifc` and `endif`, and between `else` and `endif`.

9.4.10.4 Operation

```
if ( src == true )
{
  ...
}
else
{
  ...
}
endif
```

9.4.10.5 Example

```
ifb       b0
...
else
...
endif
```

9.4.11 `loop`: Start Loop Statement

9.4.11.1 Calling Format

```
loop      src
```

9.4.11.2 Operands

- `src`: Integer register

9.4.11.3 Overview

This is used together with `endloop`. It repeatedly runs the instructions between `loop` and `endloop` according to the content of the integer register specified by `src`. The integer register comprises a loop count, an initial value for the loop-counter register, and an amount by which to increment the loop-counter register, stored in the `count`, `init`, and `step` components, respectively. (The integer register is configured via a `defi` instruction, or as uniform. For more information, see “defi: Define integer constants” and “bind_symbol (symbol_name , start_index [, end_index]).”)

The loop-counter register (`aL`) is initialized when the `loop` instruction is executed. When control reaches `endloop`, the loop-counter register is incremented by the increment amount and control returns to the `loop` instruction. After this process repeats one time more than the loop count in the integer register, the `loop` instruction ends and the next instruction after `endloop` is executed. Up to four `loop` instructions can be nested. If the loop-counter register is incremented by a negative number, its value could become 0 or less but is actually set to 255 by an underflow. Operations are undefined if a floating-point constant register is offset by a value of 96 or greater. You must denote at least one instruction between `loop` and `endloop`.

9.4.11.4 Operation

```
for ( int i = 0, aL = src.init ; i < src.count+1 ; i++, aL += src.step )
{
  ...
}
```

9.4.11.5 Example

```
defi      i0 , 10 , 0 , 1

loop      i0
add       r0 , r0 , c0[ aL ]     // Adds the total value of c0-c10 to r0
endloop
```

9.4.12 `endloop`: End Loop Statement

9.4.12.1 Calling Format

```
endloop
```

9.4.12.2 Operands

None

9.4.12.3 Overview

Specifies the location to end a loop controlled by the `loop` instruction. You must denote at least one instruction between `loop` and `endloop`.

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9.4.12.4 Operation

```
for ( int i = 0, aL = src.init ; i < src.count+1 ; i++, aL += src.step )
{
  ...
}
```

9.4.12.5 Example

```
defi      i0 , 10 , 0 , 1

loop      i0
add       r0 , r0 , c0[ aL ]     // Adds the total value of c0-c9 to r0
endloop
```

9.4.13 `breakc`: Break from Loop Statement by Condition

9.4.13.1 Calling Format

```
breakc            status0 , status1 , mode
```

9.4.13.2 Operands

- `status0`: Value (either 0 or 1) of status register 0
- `status1`: Value (either 0 or 1) of status register 1
- `mode`: Conditional mode
  - 0: OR
  - 1: AND
  - 2: Only status register 0
  - 3: Only status register 1

9.4.13.3 Overview

Forcibly exits a loop control block based on the status register values.

The equality of the values specified by `status0` (or `status1`) and status register 0 (or 1) is taken to be the condition. This condition is `true` when either status register 0 or 1 match when `mode` is 0; when both status registers match when `mode` is 1; when status register 0 matches when `mode` is 2; and when status register 1 matches when `mode` is 3.

When the condition is `true`, control is forced to exit a loop started by a `loop` instruction and then jumps to the next instruction following `endloop`.

The `breakc` instruction cannot be called between an `ifb(ifc)-endif` that is between a `loop-endloop`. The `breakc` instruction also cannot be called immediately prior to an `endloop` instruction.

9.4.13.4 Operation

```
for ( int i = 0, aL = src.init ; i < src.count+1 ; i++, aL += src.step )
{
```

```
  ...

  switch ( mode )
  {
  case 0 :
          if ( status0 == Status_register0
                   || status1 == Status_register1 )
          condition = true;
          break;
  case 1 :
          if ( status0 == Status_register0
                   && status1 == Status_register1 )
                  condition = true;
          break;
  case 2 :
          if ( status0 == Status_register0 )
                  condition = true;
          break;
  case 3 :
          if ( status1 == Status_register1 )
                  condition = true;
          break;
  }
  if ( condition == true )
          break;
}
```

9.4.13.5 Example

```
defi      i0 , 10 , 0 , 1

loop      i0
add       r0 , r0 , c0[ aL ]      // Adds the total value of c0-c9 to r0
breakc    1, 0, 2                 // Break if status register 0 is equal to 1
nop                               // Cannot issue breakc right before endloop
endloop
```

9.4.14 `cmp`: Compare

9.4.14.1 Calling Format

```
cmp       mode0 , mode1 , src0 , src1
```


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9.4.14.2 Operands

- `mode0`: Comparison mode 0
  - 0: ==
  - 1: !=
  - 2: <
  - 3: <=
  - 4: >
  - 5: >=
- `mode1`: Comparison mode 1
  - 0: ==
  - 1: !=
  - 2: <
  - 3: <=
  - 4: >
  - 5: >=
- `src0`: Temporary register, input register, or floating-point constant register
- `src1`: Temporary register, input register, or floating-point constant register

9.4.14.3 Overview

Compares the content of registers `src0` and `src1` and stores the result in the status registers.

Only the x and y components (after swizzling) of `src0` and `src1` are compared. Status register 0 stores the result of comparing the x components with the condition specified by `mode0`. Status register 1 stores the result of comparing the y components with the condition specified by `mode1`.

A status register is set to 1 if the following comparison results are `true` and 0 if they are `false`.

- `src0 == src1` when `mode0` (or `mode1`) is 0
- `src0 != src1` when `mode0` (or `mode1`) is 1
- `src0 < src1` when `mode0` (or `mode1`) is 2
- `src0 <= src1` when `mode0` (or `mode1`) is 3
- `src0 > src1` when `mode0` (or `mode1`) is 4
- `src0 >= src1` when `mode0` (or `mode1`) is 5

Both status register 0 and 1 are always updated. You cannot update only one of them.

You cannot specify a floating-point constant register for both `src0` and `src1`. You cannot specify input registers with different indices in `src0` and `src1` at the same time.

Use this instruction to distinguish between ∞, -∞, and `NaN`. A value of `0x7f0000` indicates ∞, `0xff0000` indicates -∞, and any value greater than `0x7f0000` or less than `0xff0000` indicates `NaN`.

9.4.14.4 Operation

```
switch ( mode0 )
{
```

```
case 0: status_register0 = ( src0.x == src1.x ) ? 1 : 0 ; break ;
case 1: status_register0 = ( src0.x != src1.x ) ? 1 : 0 ; break ;
case 2: status_register0 = ( src0.x < src1.x ) ? 1 : 0 ; break ;
case 3: status_register0 = ( src0.x <= src1.x ) ? 1 : 0 ; break ;
case 4: status_register0 = ( src0.x > src1.x ) ? 1 : 0 ; break ;
case 5: status_register0 = ( src0.x >= src1.x ) ? 1 : 0 ; break ;
}
switch ( mode1 )
{
case 0: status_register1 = ( src0.y == src1.y ) ? 1 : 0 ; break ;
case 1: status_register1 = ( src0.y != src1.y ) ? 1 : 0 ; break ;
case 2: status_register1 = ( src0.y < src1.y ) ? 1 : 0 ; break ;
case 3: status_register1 = ( src0.y <= src1.y ) ? 1 : 0 ; break ;
case 4: status_register1 = ( src0.y > src1.y ) ? 1 : 0 ; break ;
case 5: status_register1 = ( src0.y >= src1.y ) ? 1 : 0 ; break ;
}
```

9.4.14.5 Example

```
def       c0 , 0, 1, 2, 3

mov       r0, c0
cmp       0 , 0 , r0, c0
ifc       1 , 1 , 2
   // This is executed when r0.x == 0
endif
```

9.4.15 `end`: End Process

9.4.15.1 Calling Format

```
end
```

9.4.15.2 Operands

None

9.4.15.3 Overview

Ends vertex shader processing. This instruction must be called after a vertex shader has finished writing all output data. If this instruction is called before all output data has finished being written, the output data is undefined. This instruction uses two instructions' worth of the program size.

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10 Debug Build

Specifying the `-debug` option to `ctr_VertexShaderAssembler32.exe` results in a debug build. Although you can use the shader debugger to debug assembler objects created by a debug build, the objects may run more slowly in POD. If the `-debug` option is specified to `ctr_VertexShaderLinker32.exe`, all linked assembler objects are forced to use a debug build. If the `-nodebug` option is specified to `ctr_VertexShaderLinker32.exe`, all linked assembler objects are forced to not use a debug build.

When you link assembler objects created by a debug build with ones that are not, each of the main objects uses a debug build when it references at least one assembler object that does.

When you do not specify the `-debug` option to `ctr_VertexShaderAssembler32.exe`, the input file path is removed from the object file. Without the full input file path, the shader debugger is sometimes unable to find the source file.

11 Map Files

11.1 Overview

Specifying the -M option to `ctr_VertexShaderLinker32.exe` when linking executable files causes files to output with information about the executable files. These files are called *map files*. A map file is created with the same name as its executable file but uses the extension `.map`.

The following information is generated: *loading objects order*, *image sizes*, and *object information*. The following sections provide more details.

11.2 Loading Objects Order

This item shows the order in which the main objects were linked, which is the same as the order specified as an argument to `ctr_VertexShaderLinker32.exe`. The indicated object order is the same as the individual reference points for shader objects specified with the `glShaderBinary` function. This also shows which main objects use a debug build.

11.3 Image Sizes

This item shows the data size of the linked object files. `Instruction` indicates the number of assembly code instructions and `Swizzle` indicates the number of swizzling and masking patterns. (For more information, see section 12.3 Pattern Counts for Swizzling and Masking.) `Total` indicates the entire size after linking.

11.4 Program Code Information

This item shows the location used to store the program code in the executable file. `Program code offset` is the offset (in bytes) from the start of the executable file to the address at which the program code's data is stored. `Program code size` is the number of bytes of program code's data.

11.5 Object Information

This item shows individual symbol information, output data attribute information, and the starting program address for the linked main objects. These settings are specified using `#pragma bind_symbol` and `#pragma output_map`, and the program address is set by the `main` label, respectively.

11.6 Swizzle Pattern Data

This item shows the swizzle pattern data in the executable file.

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12 Cautions and Restrictions

Vertex shaders have the following limitations due to characteristics of the hardware.

12.1 Starting and Ending a Shader

A shader starts processing from the `main` label. If all components (x, y, z, and w) are written to the output registers specified by `#pragma output map` and the `end` instruction is called, processing ends and then restarts for the next vertex. The shader does not run properly if values are not written to the registers specified by `#pragma output map`. You must explicitly call the `end` instruction at the end of processing.

After data has been written to every output register, all required processing is recognized to be complete. It is uncertain whether instructions will run after the last instruction that writes to the output registers. Normal operations might not result when calling an instruction to read or write registers after the last instruction to write to the output registers. Do not call any instructions other than `nop` between the instruction for the last write to output registers and the `end` instruction.

You can only write to an output register once. Operations are not guaranteed if you write to an output register more than once. This also applies to writing to each component.

If you don't read an input register at least once during processing of a single vertex, the shader might not work normally. Be sure to execute at least one read instruction on at least one component of any input register.

12.2 Step Count

Programs have a maximum step count of 512.

The `def`, `defi`, `defb`, `ret`, `else`, `endif`, and `endloop` instructions are not calculated as step counts.

12.3 Pattern Counts for Swizzling and Masking

There is a maximum number of patterns for the combination of masking, replacing (swizzling), and adding signs to input components. This upper limit is 128, of which no more than 32 patterns are usable with the `mad` instruction.

Code 12-1 Pattern Count Example 1

```
add       r0 , r1.xy , -r2.zw
add       r2 , r0.x, r3
mul       r3 , r2.xy , -r3.zw
add       r4, r2.xxxx, r5.xyzw
```

The first and third instructions in this example have the same pattern. The second and fourth instructions also have the same pattern, so the combined pattern count is 2.

Code 12-2 Pattern Count Example 2

```
add       r0 , r1.xy , -r2.zw
mul       r2.xy , r0.x , c2.y
mad       r3 , r2.xy , -r3.zw , r1.w
cmp       0 , 1 , r1.x , c0.y
```

The first and third instructions in this example are considered to have the same pattern because the combination for the third instruction (`mad`) is the same as the combination for the first instruction (`add`) except for the `src2` operand. This pattern is also treated as being usable by the `mad` instruction.

The combination of `src0` and `src1` in the fourth instruction (`cmp`) is the same as the combination of `src0` and `src1` in the second instruction (`mul`), so the second and fourth instructions are considered to have the same pattern.

12.4 Control Instruction Limitations

You must call an ending control instruction after a starting control instruction. Specifically, you must use the following combinations.

- `ifb (-else)-endif`
- `ifc (-else)-endif`
- `call-ret`
- `callb-ret`
- `callc-ret`
- `loop-endloop`

It is illegal to jump outside of these control blocks using the `jpc` or `jpb` jump instructions. You also cannot call `ret` from within an `if` or `loop` block.

When `call` instructions are nested, operations are undefined when another `call` instruction is called immediately before a `ret` instruction within a subroutine.

You cannot use the `jpc` or `jpb` instruction between the `main` and `endmain` labels to jump to an external location. You also cannot call the instructions from a subroutine to jump to an external location. You cannot jump to a `ret` instruction within a subroutine. Operations are undefined for all of these controls.

12.5 Instructions That Cannot Be Called Consecutively

Some instructions cannot be called consecutively. There are also certain combinations of instructions that cannot be called consecutively.

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12.5.1 Consecutive Calls of else/endif/ret/endloop

The `else`, `endif`, `ret`, and `endloop` instructions cannot be called consecutively.

Code 12-3 Instructions That Cannot Be Called Consecutively

```
ifb   b0
  nop
  nop
  ifb   b1
    nop
  else
    nop
    nop
  endif               // Error
else
  nop
  nop
  call   subroutine    // This causes an error because ret is called at the end
                      // of the call destination
endif
```

12.5.2 Consecutive Calls of mova

The `mova` instruction also cannot be called consecutively.

12.5.3 Calls of jpc/jpb Immediately Before else/endif/ret/endloop

`jpc` and `jpb` cannot be called immediately before `else`, `endif`, `ret`, or `endloop`.

Code 12-4 jpc and jpb Cannot Be Called Immediately Before else, endif, ret, or endloop

```
ifb       b0
  nop
  nop
jumplabel:
  nop
  nop
  jpb     b1, true, jumplabel    // Prohibit to call right before endif
endif
```

12.5.4 Calling breakc Before Endloop

`breakc` cannot be called immediately before `endloop`.

Code 12-5 breakc Cannot Be Called Immediately Before endloop

```
loop      i0
  nop
  breakc 1, 1, 1          // It's prohibited to call breakc right before endloop
endloop
```

12.6 Registers That Cannot Be Used Simultaneously

In general, you cannot specify two or more floating-point constant registers for assembly code instructions that specify two or more `src` operands. You also cannot specify two or more input registers. However, you can specify any number of input registers if they have the same index.

Code 12-6 Registers That Cannot Be Used Simultaneously

```
add  r0 , c0 , c0         // Error
add  r0 , c0 , c1         // Error
add  r0 , v0 , v0         // This is not an error.
add  r0 , v0 ,v1          // Error
```

Macro instructions are checked for errors after they have been expanded.

12.7 Instruction Latency

The following table shows the latency of running each instruction.

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Table 12-1 Instruction Latency

Instruction	Latency (in clock cycles)
`add`	3
`dp3`	5
`dp4`	5
`dph`	5
`dst`	3
`exp`	4
`flr`	2
`litp`	4
`log`	4
`mad`	4
`max`	2
`min`	2
`mov`	2
`mova`	4
`mul`	3
`nop`	1
`rcp`	4
`rsq`	4
`sge`	2
`slt`	2
`cmp`	4
Other branch instructions	1 to 3

12.7.1 Arithmetic and cmp Instruction Latency

Although the previous table gives the approximate number of clock cycles for the latency of arithmetic instructions and the `cmp` instruction, these values may change depending on the preceding and following instructions. Latency may be reduced by queuing up instructions that use unrelated registers for their calculations.

The various arithmetic instructions have stages that are equal to the number of latency cycles in the table above, plus 1 cycle. There is a read stage (for reading the register), an execution stage (for performing the computation), an instruction-specific post-process stage, and a write stage (for writing

the result to the register). For example, the `add` instruction has four stages: a read stage, a computation stage, a post-process stage, and a write stage. (Some instructions do not have post-process stages).

Basically speaking, the final stage of an arithmetic instruction is the write stage. If the result of one instruction is used by the next instruction, the write stage (the final stage of the prior instruction) and the read stage (the first stage of the next instruction) can execute at the same time. (The read stage of the next instruction does not have to wait for the write stage of the prior instruction to end.) For this reason, one cycle can be eliminated, and the number of stages for the instruction = (the number of latency cycles + 1).

12.7.2 Branch Instruction Latency

The previous table gives a latency of 1 to 3 for branch instructions. The latency of the branch instructions varies, depending on how the program counter changes after the branch instruction.

If the program counter is incremented by 1 after the execution of the branch instruction, the latency is 1. If the program counter is incremented by 2, then the latency is 2. If the program counter is changed in some other way, the latency is 3.

12.7.3 Output Order of Calculation Results

Instruction results never affect the results of an earlier instruction. This behavior is guaranteed and never causes any stalls, because register reads for earlier instructions are always executed before register reads for later instructions, regardless of the latency lengths of the earlier and later instructions.

For example, when a later instruction writes to the source register for an earlier instruction, the results of the later instruction's write are never used as input for the earlier instruction.

Code 12-7 Output Order of Calculation Results, Example 1

```
exp  r0 , r1.x
mov  r1 , c0
```

This example uses the source register of a high-latency earlier instruction as the destination register of a low-latency later instruction. (`exp` takes four clock ticks, while `mov` takes only two.) Running this kind of code never results in the output result of `mov` being used as the input for `exp`. (The results of `mov` have no affect on the `exp` calculation.)

Also, calculation results are guaranteed to be output in the order of instruction execution.

Code 12-8 Output Order of Calculation Results, Example 2

```
exp  r0 , r1.x
mov  r0 , c0
```


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This example uses the same destination register for both a high-latency earlier instruction and a low-latency later instruction. Running this kind of code never causes the result of the high-latency `exp` instruction to be output later than the result of the low-latency `mov` instruction.

In cases like this one where there are register dependencies, the earlier instruction's write is canceled as soon as the later instruction is decoded, guaranteeing stall-free operation. For each individual component, the destination registers of earlier and later instructions are compared; and if the instructions write to the same register, the write is canceled in the earlier instruction. In other words, writes are only canceled for an earlier instruction when a later instruction writes to the same register component.

Successive writes to the same register do not cause stalls, regardless of whether the successive writes are to the same register component.

Code 12-9 Output Order of Calculation Results, Example 3

```
exp  r0.x , r1.x
mov  r0.y , c0
```

This example includes successive writes to `r0`, one to `r0.x` and one to `r0.y`, so there are no successive writes to the same component. The `exp` write to `r0.x` is not canceled, but no stall occurs.

Code 12-10 Output Order of Calculation Results, Example 4

```
exp  r0.xyz , r1.x
mov  r0.xyzw , c0
```

In this example, the `exp` write to `r0.xyz` is canceled. No stall occurs.

12.7.4 Stalls Caused by Conflicts When Outputting Calculation Results

If a low-latency instruction is executed after a high-latency instruction and they both complete simultaneously, the results of the low-latency instruction are output after one clock cycle. Multiple registers are never written simultaneously.

Code 12-11 Simultaneous Instruction Completion

```
exp  r0 , r1.x
mul  r2 , c3 , r4
```

When this code is run, `exp` takes four clock cycles and `mul` takes three clock cycles, such that the results of both `r0` and `r2` would be output simultaneously. However, to avoid multiple register writes at once, the output of `r2` is delayed by one clock cycle.

12.7.5 Stalls Caused by Conflicts Among Arithmetic Units

The `mad`, `dp3`, `dp4`, `dph`, and `add` instructions all require use of the adder. If these instructions are executed in sequence, the adder pipeline may get backed up, with instructions executed later waiting for previously executed instructions to finish, causing higher latency.

The adder is used during the first cycle of the `add` instruction, the second cycle of the `mad` instruction, and the second and third cycles of the `dp3`, `dp4`, and `dph` instructions.

However, calling the `dp3`, `dp4`, and `dph` instructions in sequence is not usual, and calling them in sequence still does not cause any stall due to conflicts among the arithmetic units. Neither calling just `dp3`, just `dp4`, or just `dph` in succession, nor calling all three successively in combination causes a stall due to conflicts among the arithmetic units.

12.7.6 Stalls Caused by Instruction Dependencies

Dependencies between issued instructions may cause stalls. This phenomenon occurs when an instruction stores its calculated results in a register that is used as a source register by the next instruction, as shown in the following code.

Code 12-12 Register Use Causing a Stall

```
add r0, r1, r2
mul r4, r0, r3
```

When run, the preceding code stalls because the results output to `r0` are used by the very next instruction as a source register.

Operation stalls even if the two instructions use different components of the same register.

Code 12-13 Using Different Register Components Still Causes a Stall

```
add r0.x, r1, r2
mul r4, r0.y, r3
```

In this code, although the latter instruction does not reference the `r0.x` component to which the earlier instruction outputs its result, the latter instruction does reference the `r0` register (as `r0.y`), thereby causing a stall.

When there are multiple successive writes to the same register, the write performed by the first instruction is canceled. (For more information, see section 12.7.3 Output Order of Calculation Results.) However, the canceled instruction can still sometimes cause later instructions (those that read the written results) to stall.

Code 12-14 Cancelled Instruction Causing a Stall

```
dp4 r0.x, r1, r2
mov r0.x, r1
mul r4, r0, r3
```


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This code uses the same destination register for the `dp4` and `mov` instructions, so the `dp4` write is canceled. However, the later `mul` instruction depends on both the `dp4` and `mov` instructions, and therefore stalls. Of the two instructions preceding `mul`, `dp4` has greater latency than `mov`, so this situation results in `mul` stalling until the `dp4` instruction is done executing.

12.7.7 Unconditional Stalls

Calling the `mova` instruction unconditionally causes a stall lasting three clock cycles. Unlike stalls caused by instruction dependencies, calling `mova` always causes a stall. You cannot avoid a stall by inserting non-dependent instructions (instructions using registers not affected by preceding or succeeding instructions) between the `mova` instruction and the instruction that reads the address register written to by the `mova` instruction.

Code 12-15 Unconditional Stall Caused by a mova Instruction

```
mova a0.x, r0
nop
nop
nop
mov r1, c[a0.x]
```

This code inserts three `nop` instructions between the `mova` instruction and the `mov` instruction that reads the address register written to by the `mova` instruction. However, the `mova` instruction causes a stall regardless of whether the `nop` instructions are inserted.

12.8 Results of Exceptional Operations

The vertex shader exhibits the following behavior as the result of calculating an exceptional operation.

- `NaN` is output for the logarithm of a negative value or -0.
- `NaN` is output for the square root of a negative value or -0.
- `NaN` is output for operations that use `NaN` as an input value (except for the `cmp` instruction).
- Negative infinity (-∞) is output when infinity (∞) is subtracted from a number.
- Negative infinity (-∞) is output for the logarithm of a non-normalized number (a number with an exponent of 0 and a nonzero significand) or +0.
- Infinity (∞) is output when there is an overflow.
- Negative infinity (-∞) is output when there is an underflow.
- Infinity (∞) or negative infinity (-∞) is output for a division by positive or negative 0.
- When the `rcp`, `rsq`, `exp`, or `log` instruction performs calculations that result in ∞ or -∞, sometimes `NaN` is output instead.

Use a `cmp` instruction if you need to distinguish between ∞, -∞, and `NaN`. For more information, see section 9.4.14 `cmp`: Compare.

12.9 Limitations Related to Invalid Data Output

Operations are not guaranteed when a vertex attribute value of NaN (Not a Number) is output from the vertex shader (written to an output register). Do not give NaN as an input vertex attribute or uniform value from the application, or as the calculation result output from a vertex shader.

12.10 Shader Implementations That Cause Invalid Operations

The order in which shader assembler instructions are executed can cause the hardware to carry out invalid operations. This can cause the GPU to hang. The shader assembler or linker displays warnings if a shader is implemented in a way that could cause invalid operations.

The following sections describe the kinds of conditions that could cause invalid operations.

12.10.1 Invalid Operation Due to a `mova` Instruction

Invalid operations can be caused by any number of conditions. This section describes invalid operations specifically caused by a `mova` instruction.

Some of these conditions are understood completely, while other conditions are highly suspect. Use the provided information to avoid known issues and troubleshoot current problems, but understand that your implementation of other portions of the shader may introduce results not previously seen. Factors to consider include the types of instructions preceding what appears to be the offending instruction, the combination of registers used, and branch instructions in registers. Factors to ignore include execution timing and register contents other than branch instructions. Ultimately, a successful shader implementation is determined by trouble-free operations as exhibited on a production unit.

12.10.1.1 Executing a `mova` Instruction as the Second-to-Last Instruction

Shaders that execute a `mova` instruction as the second-to-last instruction with no dependence on the final instruction can cause the hardware to hang. For vertex shaders, the last instruction executed is the last one written to the register. For geometry shaders, the last instruction specifies an `end` instruction.

Code 12-16 Vertex Shader Causing Invalid Operations

```
// Vertex Shader 1
mova a0.x, r0     // Final instruction right after mova.
mov o0, r2        // Final instruction does not use a0.x, raising risk of a hang.
end
```

Code 12-17 Vertex Shader Not Causing Invalid Operations

```
// Vertex Shader 2
mova a0.x, r0     // Final instruction right after mova.
mov o0, c[a0.x]   // However, final instruction does use a0.x, so no problem.
end
```


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Code 12-18 Geometry Shader Causing Invalid Operations

```
// Geometry Shader
mova a0.x, r0     // Execute mova.
end               // End right after mova raises risk of a hang.
```

The shader assembler outputs warning `400a0003` whenever a `mova` instruction comes just before the `end` instruction. The shader assembler outputs warning `400a0004` whenever a `mova` instruction is followed by an instruction to write to an output register and then by an `end` instruction, and the register written to by the `mova` instruction is not used by the next instruction. Avoid this set of circumstances by taking steps such as inserting a `nop` instruction just before the final instruction, or changing the order of instructions.

12.10.1.2 Executing `mova` Instructions Just Before and Just After Certain Other Instructions

Combinations of the `mova` instruction with the `else-endif`, `call-ret`, and `loop-endloop` instructions can cause the hardware to hang. This section describes all three combinations.

The hardware might hang for an `ifb` or `ifc` code clause when there is a `mova` instruction both right before the `else` and right after the `endif`. The shader assembler outputs warning `40070003` when it detects such an implementation. You can avoid this sequence by taking steps such as inserting a `nop` instruction just before the final instruction or changing the order of instructions.

Code 12-19 else-endif Clause Causing Invalid Operations

```
ifb b0
   ...
mova a0.x, r0    // Executes a mova just before the else. Jumps to just after the
                 // endif after executing.
else
   ...
endif
mova a0.x, r1     // This can cause a hang when jumping from just before the else.
```

The hardware might hang for a `call`, `callb`, or `callc` code clause when there is a `mova` instruction both right before the final `ret` of the called subroutine and right after the subroutine returns. The shader assembler outputs warning `4009000c` when it detects such an implementation. You can avoid this sequence by taking steps such as inserting a `nop` instruction just before the final instruction or changing the order of instructions.

Code 12-20 call-ret Clause Causing Invalid Operations

```
main:             // main function.
...
call l_function // Jump to l_function.
mova a0.x, r0     // mova again right after returning from l_function.
...
```

```
end                // main function ends.

l_function:        // Subroutine.
...
mova a0.x, r1      // mova just before ret.
ret
```

The hardware might hang when there is a `mova` instruction both right after a `loop` instruction and right before an `endloop` instruction. The shader assembler outputs warning `40070004` when it detects such an implementation. You can avoid this sequence by taking steps such as inserting a `nop` instruction just before the final instruction or changing the order of instructions.

Code 12-21 loop-endloop Clause Causing Invalid Operations

```
loop i0
   mova a0.y, r0 // mova right after loop.
   ...
   mova a0.x, r1 // mova right before endloop.
endloop
```

12.10.1.3 Stalling on a `mova` Instruction and Branching Right Afterwards

The hardware might hang when executing a `mova` instruction that a) stalls caused by a dependence on the preceding instruction and b) is immediately followed by a branch instruction. Branch instructions are `jpb`, `jpc`, `call`, `callb`, `callc`, `ifb`, `ifc`, and `breakc`.

Code 12-22 mova Followed by a Branch Instruction Causing Invalid Operations

```
dp4 r0, r1, r2            // Write to r0.
mova a0.x, r0.x           // r0 depends on dp4, so this stalls.
call l_function           // Branch instruction right after a stalling mova.
```

The shader assembler outputs warning `400a0005` when it detects such an implementation (a branch instruction following a `mova` instruction that has a temporary register as its source). You can avoid this sequence by taking steps such as inserting a `nop` instruction just before the final instruction or changing the order of instructions.

12.10.1.4 Invalid Operations Caused by Instruction Register Dependencies Before or After a `mova` Instruction

The GPU stops responding if all of the following conditions are met for certain `mova` instructions:

- Condition 1: The instruction immediately following `mova` is to read a register that was written by an instruction prior to `mova`. This dependency causes the instruction immediately after `mova` to stall for 1 clock. In such cases, the instruction is ignored as a 3-clock unconditional `mova` stall.

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- Condition 2: `mova` or either of the two instructions after `mova` read an input register, or they read a temporary register. In such cases, the register index and components are ignored.
- Condition 3: `mova` has no dependencies with the preceding instruction and `mova` does not stall because of register dependencies.

The following example shows an occurrence of this phenomenon.

Example:

```
mul       r0, c0, r1      // Stall the instruction immediately after mova.
// (Condition 1.)
mova      a0.x, v0.x      // Stall does not occur because of dependencies on
// the previous instruction. (Condition 3.)
add       r0, r0, c1      // 1-clock stall because of dependencies on
// instructions prior to mova. (Condition 1.)
mov       r5, v1          // Reading the same input register as mova. (Condition
2.)
```

To detect this phenomenon, a warning (400a0006) is output from the shader assembler when the following conditions are met:

- Warning condition 1: When an instruction immediately after `mova` reads a temporary register (a required condition for the occurrence of Condition 1 for this malfunction).
- Warning condition 2: `mova` or either of the two instructions after `mova` reads an input register or if any of them read a temporary register (Condition 2 for this malfunction).

Checks are not made for instruction stalls. If a warning appears, insert a `nop` instruction and change the order of the instructions to work around the conditions that cause this malfunction.

12.10.2 Invalid Operation Due to a Specific Order of Instructions

The hardware carries out invalid operations when four consecutive instructions meet all of the following conditions. These conditions are *necessary and sufficient* to cause invalid operation. (It cannot be caused any other way, and is always caused if these conditions are met.) If all these conditions are met, the GPU always hangs.

1. The first and third instructions are latency-2 instructions (`flr`, `litp`, `max`, `min`, `mov`, `sge`, `slt`, `abs`).
2. The second instruction is a latency-2 or lower instruction (`flr`, `litp`, `max`, `min`, `mov`, `sge`, `slt`, `abs`, `nop`).
3. The fourth instruction is a branch instruction (`jpb`, `jpc`, `call`, `callb`, `callc`, `ifb`, `ifc`, `breakc`).
4. The first instruction stalls. This stall must be for at least two clock cycles when the second instruction is `nop`, and at least three clock cycles when the second instruction is not `nop`.
5. The `dst` of the first instruction is not the same register as any `src` of the second instruction.

6. The `dst` of the second instruction is not the same register as any `src` of the third instruction.

7. The `dst` of the first instruction is the same register as a `src` of the third instruction.

8. The `dst` of the third instruction is the same register as a `src` of the third instruction.

The phrase “the same register” in conditions 5, 6, 7, and 8 above describes when the `dst` and `src` operands refer to a register having the same index and the same type, regardless of the component specified by the operand. In other words, `r0` and `r0` are the same register, and likewise `r1.x` and `r1.y` are the same register.

The following code shows an example of this situation.

Code 12-23 Instruction Ordering That Causes Invalid Operations

```
rcp r1, r2.x      // This instruction causes the first instruction to stall.
min r0, r1, r2    // First instruction of the four that meet these conditions.
max r3, r4, r5    // Second instruction.
slt r5, r0, r5    // Third instruction.
call l_function   // Fourth instruction.
```

The first instruction of this chain is `min` and the third is `slt`, both latency-2 instructions meeting condition 1. The second instruction is `max`, also latency-2 and thus fulfilling condition 2. The fourth instruction is `call`, meeting condition 3. The `min` instruction waits for register `r1` to be written to by the `rcp` instruction, causing a stall for three clock cycles and fulfilling condition 4. The `min` instruction’s `dst` and the `max` instruction’s `src` are not the same, meeting condition 5. The `max` instruction’s `dst` and the `slt` instruction’s `src` are not the same, meeting condition 6. The `min` instruction’s `dst` and the `slt` instruction’s `src` are the same, meeting condition 7. Last, the `slt` instruction has the same `dst` and `src1` operands, meeting condition 8.

The shader assembler outputs warning `400a0001` or `400a0002` when it detects an implementation that meets all these conditions except condition 4.

The shader assembler has no means of telling whether the first instruction stalls, and so cannot evaluate condition 4. You can get a rough estimate of the length of any stall by using the performance checking feature of the shader linker, but this is not a conclusive evaluation. Stalling conditions depend on the instructions executed previously and how registers are used. Shaders either definitely cause invalid operations, or definitely do not cause invalid operations. A shader that operates properly even when the shader assembler outputs a warning must therefore not meet condition 4, and is thus safe to continue using. If a shader does cause invalid operations, avoid this sequence by taking steps such as inserting a `nop` instruction just before the final instruction or changing the order of instructions.

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13 Error Messages for the Assembler and Linker

13.1 Overview

This chapter describes the error messages output by the assembler and linker. Errors are output in the following format.

```
Input filename (error line number): Error level (Error Code): Error
description
```

The error level is either *warning* or *error*. Processing can continue when there is a warning. The input filename, error line number, and other information may not be displayed for some types of errors.

13.2 Assembler Error Messages

This section describes the errors and error codes output by the assembler.

80010001

```
(80010001): -O option cannot be specified more than once.
```

The `-O` option cannot be specified more than once.

80010003

```
(80010003): Definition key is not specified with -D option
```

The key to define for the `-D` option has not been specified.

Specify it in the format "`-Dkey`" or "`-Dkey=value`".

80010004

```
(80010004): Definition value is not specified with ‘argument name’ macro.
```

A value has not been correctly set for the definition macro with the `-D` option.

Specify it in the format "`-Dkey`" or "`-Dkey=value`".

80010005

```
(80010005): ‘argument name’ includes illegal character.
```

A macro was defined for the `-D` option using illegal characters.

Use single-byte alphanumeric characters and underscores for macro names.

80010006

```
(80010006): ‘Macro name’ macro is redefined.
```

More than one macro with the same name has been defined with the `-D` option.

80010007

```
(80010007): Only one assembler file can be specified as input.
```

More than one assembler file has been specified as an input file. Specify only one assembler file.

80010008

```
(80010008): Input file is not specified.
```

An assembler file has not been specified as an input file.

8001000b

```
(8001000b): ‘Macro name’ macro name cannot start from number.
```

You cannot use a number as the first character of a macro with the `-D` option.

8001000d

```
(8001000d): Unknown options are specified.
```

Unknown options were specified.

80030001

```
(80030001): Cannot open ‘filename’.
```

The specified assembler file cannot be opened.

80030002

```
(80030002): Include filename is not specified.
```

A filename has not been specified with a `#include` statement.

Specify it in the format "`#include 'filename'`".

80030003

```
(80030003): Syntax error in #include.
```

A `#include` statement was denoted incorrectly.

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Specify it in the format "`#include 'filename'`".

80030004

```
(80030004): Cannot open include file “filename”.
```

Could not open an include file.

Specify the include path using the `-I` option.

80030005

```
(80030005): Definition key is not specified.
```

A `#define` statement was denoted incorrectly.

Use the format "`#define key value`".

80030006

```
(80030006): Definition key include illegal character.
```

Use single-byte alphanumeric characters and underscores for macro names defined by `#define` statements.

80030007

```
(80030007): ‘Macro name’ macro is redefined.
```

Duplicate macros have been defined by `#define` statements.

80030008

```
(80030008): Definition key is not specified.
```

A `#undef` statement was denoted incorrectly.

Use the format "`#undef key`".

8003000b

```
(8003000b): Correspondent “#ifdef” is not found.
```

A `#endif` statement is missing a corresponding `#ifdef` statement.

8003000c

```
(8003000c): Undefined directive.
```

An unsupported preprocessor pseudo-instruction has been specified.

8003000d

```
(8003000d): #ifdef is not closed.
```

A `#ifdef` statement is missing a corresponding `#endif` statement.

8003000e

```
(8003000e): Syntax error. Macro is not specified.
```

A macro has not been specified for a `#ifdef` statement.

Specify the format "`#ifdef macro`".

8003000f

```
(8003000f): Syntax error. Invalid string is detected after macro.
```

A `#ifdef` statement was entered incorrectly.

Specify the format "`#ifdef macro`".

80030010

```
(80030010): Syntax error. Macro is not specified.
```

A macro has not been specified for a `#ifndef` statement.

Specify the format "`#ifndef macro`".

80030011

```
(80030011): Syntax error. Invalid string is detected after macro.
```

A `#ifndef` statement was entered incorrectly.

Specify the format "`#ifndef macro`".

80030012

```
(80030012): Syntax Error. Invalid string is detected after directive.
```

Use single-byte alphanumeric characters and underscores for macro names specified by `#if`, `#ifdef`, and `#ifndef` statements.

80030013

```
(80030013): Syntax Error. Invalid string is detected after directive.
```

An invalid string was detected after a `#else` statement.

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80030014

```
(80030014): Correspondent "#ifdef" is not found.
```

Could not find a `#ifdef` statement corresponding to an `#else` statement.

80030015

```
(80030015): Syntax error. Invalid expression is detected.
```

A `#if` statement was denoted incorrectly.

80030017

```
(80030017): Syntax error. Invalid expression is detected.
```

A macro entered after a `#if` statement uses invalid characters.

Use single-byte alphanumeric characters and underscores for macro names specified with `#if` statements.

80030018

```
(80030018): #error
```

This error is intentionally output by an `#error` statement.

80030019

```
(80030019): The top character of definition key must not be number.
```

You cannot use a number as the first character of a macro name defined using a `#define` statement.

8003001a

```
(8003001a): Macro parentheses have not been closed properly.
```

Parentheses are not denoted correctly in a definition using the `#define` statement.

8003001b

```
(8003001b): Invalid character is detected in macro argument.
```

One of the arguments to a function macro uses an illegal string.

Use single-byte alphanumeric characters and underscores for macro names.

8003001c

```
(8003001c): Duplicate macro argument is detected.
```

The same string is used more than once in the arguments to a function macro.

8003001d

```
(8003001d): Invalid macro argument is specified.
```

An incorrectly defined macro function was used.

8003001e

```
(8003001e): pragma command bind_symbol is invalid format.
```

The `#pragma bind_symbol` statement was denoted incorrectly.

8003001f

```
(8003001f): Undefined pragma command.
```

An unsupported `pragma` command was specified.

80030020

```
(80030020): Start index should be less than or equal to end index.
```

The starting register index is larger than the ending register index in a `#pragma bind_symbol` statement.

80030021

```
(80030021): Binding symbol name is duplicated.
```

Duplicate symbol names are defined by `#pragma bind_symbol` statements.

80030022

```
(80030022): Invalid register index is specified.
```

An invalid register index has been defined by a `#pragma bind_symbol` statement.

(The maximum number of registers was exceeded.)

80030023

```
(80030023): Specified registers are already bound to other symbol.
```

The same input register was bound to more than one symbol by `#pragma bind_symbol` statements.

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An input register corresponds to a single register and cannot be bound to more than one symbol name.

80030024

```
(80030024): Pragma command output_map is invalid format.
```

The `#pragma output_map` statement was denoted incorrectly.

80030025

```
(80030025): Invalid data name is specified for pragma command output_map.
```

An invalid data attribute name has been specified in a `#pragma output_map` statement.

8003002c

```
(8003002c): Specified register is already mapped.
```

The register specified by a `#pragma output_map` statement has already been specified by another `#pragma output_map` statement.

8003002d

```
(8003002d): Specified attribute is already mapped.
```

The data attribute name specified by a `#pragma output_map` statement has already been specified by another `#pragma output_map` statement.

80030033

```
(80030033): If all textures are mapped, texture1 and texture2 need to be mapped to
same register.
```

`texture1` and `texture2` must be mapped to the same register when all textures have been defined by `#pragma output_map` statements.

80030034

```
(80030034): comment /* */ is not closed.
```

A comment of the `/* … */` type has not been closed properly.

8003003b

```
(8003003b): Pragma command suppress_linker_warning_level2_level3 is invalid format.
```

`#pragma suppress_linker_warning_level2_level3` is coded incorrectly.

80040001

```
(80040001): No vertex shader instruction.
```

No shader instruction has been denoted.

80040005

```
(80040005): loop instruction is not closed by endloop.
```

A `loop` instruction is missing a corresponding `endloop` instruction.

80040007

```
(80040007): if or else instruction is not closed by endif.
```

An `ifc` or `ifb` instruction is missing a corresponding `endif` instruction.

80040009

```
(80040009): Unknown instruction.
```

An unknown shader instruction was denoted.

8004000c

```
(8004000c): The number of operand is short.
```

There are not enough operands.

8004000d

```
(8004000d): There are some extra operand.
```

Too many operands have been specified.

8004000e

```
(8004000e): “Operand” is unknown operand type.
```

An unknown operand type was specified.

8004000f

```
(8004000f): “Operand” is invalid format operand.
```

An invalid operand type was specified.

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80040010

```
(80040010): “Operand” is invalid offset.
```

The register offset notation is not correct.

80040011

```
(80040011): “Operand” is invalid address register offset.
```

The register offset notation for the address register is not correct.

80040012

```
(80040012): “Operand” include unknown component.
```

An unknown component has been specified.

80040015

```
(80040015): break instruction is not between loop and endloop.
```

A `break` instruction was not placed between a `loop` and `endloop` instruction.

80040016

```
(80040016): loop instruction nest achieved limit.
```

A `loop` instruction was used beyond the nesting limit.

Up to 4 `loop` statements can be nested.

80040017

```
(80040017): Correspondent loop instruction is not found.
```

An `endloop` instruction is missing a corresponding `loop` instruction.

8004001a

```
(8004001a): if instruction nest achieved limit.
```

An `ifb` or `ifc` instruction was used beyond the nesting limit.

You can nest up to eight `ifb` and `ifc` instructions.

8004001b

```
(8004001b): Correspondent if instruction is not found.
```

An `else` instruction is missing a corresponding `ifc` or `ifb` instruction.

8004001d

```
(8004001d): loop instruction is not closed, but ret instruction is called.
```

A `ret` instruction cannot be placed between a `loop` and `endloop` instruction.

8004001f

```
(8004001f): if else instruction is not closed, but ret instruction is called.
```

A `ret` instruction cannot be placed between an `ifc`, `ifb`, or `else` instruction and an `endif` instruction.

80040021

```
(80040021): “Operand” is invalid format operand.
```

An invalid operand format was specified.

80040022

```
(80040022): “Operand” is invalid index.
```

An invalid index was specified for an operand register.

80040023

```
(80040023): “Operand” is invalid format operand.
```

The parentheses used to specify an offset for an operand register have not been closed.

80040024

```
(80040024): “Operand” is invalid offset.
```

An invalid offset has been specified for an operand register.

80040025

```
(80040025): “Operand” is invalid offset.
```

A register specified with an offset for an operand register is not allowed to use an index.

80040026

```
(80040026): “Operand” is invalid offset.
```


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An invalid index was specified for an operand register.

80040027

```
(80040027): Correspondent if instruction is not found.
```

An `endif` instruction is missing a corresponding `ifc` or `ifb` instruction.

8004002a

```
(8004002a): Const register definition is duplicate.
```

Duplicate floating-point constant registers have been defined by `def` instructions.

8004002b

```
(8004002b): Bool register definition is duplicate.
```

Duplicate Boolean registers have been defined by `defb` instructions.

8004002c

```
(8004002c): Integer register definition is duplicate.
```

Duplicate integer registers have been defined by `defi` instructions.

80040031

```
(80040031): “Label name” is already used label name.
```

Duplicate label names have been used.

80040032

```
(80040032): Invalid label name is specified.
```

An invalid character has been used in a label name.

Use single-byte alphanumeric characters and underscores for label names.

80040035

```
(80040035): Error occurred while replacing macro instruction.
```

An error occurred while expanding a macro instruction.

Check whether you have already used a `def` instruction to define the floating-point constant registers that are automatically defined by `sincos` and other instructions.

80040039

```
(80040039): Cannot break from if statement.
```

You cannot place a `break` instruction within an `ifc` or `ifb` control block.

8004003a

```
(8004003a): ret instruction cannot be used just after endloop or endif.
```

A `ret` instruction cannot be called immediately after an `endloop` or `endif` instruction.

8004003b

```
(8004003b):At least 1 instruction need to be between if and else.
```

At least one assembly code instruction is required between an `ifb` or `ifc` instruction and an `else` instruction.

8004003c

```
(8004003c):At least 1 instruction need to be between else and endif.
```

At least one assembly code instruction is required between an `else` instruction and an `endif` instruction.

8004003d

```
(8004003d):def instruction cannot specify the register defined by pragma
bind_symbol.
```

A register specified with `#pragma bind_symbol` cannot be specified with a `def` instruction.

8004003e

```
(8004003e):defb instruction cannot specify the register defined by pragma
bind_symbol.
```

A register specified with `#pragma bind_symbol` cannot be specified with a `defb` instruction.

8004003f

```
(8004003f):defi instruction cannot specify the register defined by pragma
bind_symbol.
```

A register specified with `#pragma bind_symbol` cannot be specified with a `defi` instruction.

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80040040

```
(80040040):At least 1 instruction need to be between loop and endloop.
```

At least one assembly code instruction is required between a `loop` instruction and an `endloop` instruction.

80040041

```
(80040041):mova cannot be called continuously.
```

The `mova` instruction cannot be called consecutively.

80040042

```
(80040042): ret cannot be called just after jpb and jpc.
```

The `ret` instruction cannot be called immediately after the `jpb` or `jpc` instruction.

80050001

```
(80050001): Cannot open output file.
```

Cannot open the output file.

Confirm whether there is another file with the same name and read-only or other attributes.

80050003

```
(80050003): The size of swizzle register is short.
```

The maximum number of swizzling and masking patterns has been exceeded.

80050005

```
(80050005): ret instruction cannot be found for Label “label name”.
```

A label is missing a corresponding `ret` instruction.

A `ret` instruction is required for labels that are called as subroutines.

80050007

```
(80050007): The number of label is too big.
```

You cannot configure more than 65,535 labels.

8005000a

```
(8005000a): The exceptional jump is detected.
```

Exceptional jump control has occurred.

For more information, see section 12.5 Instructions That Cannot Be Called Consecutively.

8005000b

```
(8005000b): Cannot jump out from if statement and loop statement.
```

A `jpb` or `jpc` instruction cannot be used to jump from within an `ifc` or `ifb` instruction and an `endif` instruction, or from within a `loop` instruction and an `endloop` instruction, to outside a control block.

8005000c

```
(8005000c): breakc cannot be called just before endloop instruction.
```

A breakc instruction cannot be called immediately before an endloop instruction.

8005000d

```
(8005000d): jpb and jpc cannot be called just before endloop instruction.
```

A `jpb` or `jpc` instruction cannot be called immediately before an `endloop` instruction.

8005000e

```
(8005000e): jpb and jpc cannot be called just before endif instruction.
```

A `jpb` or `jpc` instruction cannot be called immediately before an `endif` instruction.

8005000f

```
(8005000f): jpb and jpc cannot be called just before else instruction.
```

A `jpb` or `jpc` instruction cannot be called immediately before an `else` instruction.

80050010

```
(80050010): jpb and jpc cannot jump into if statement and loop statement.
```

A `jpb` or `jpc` instruction cannot jump into an `if-endif or loop-endloop` statement from an external location.

80060004

```
(80060004): “Operand” is invalid operand type.
```

An unsupported operand was specified for a shader instruction.

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80060005

```
(80060005): Value cannot be specified for “operand”.
```

You cannot specify a direct value for an operand.

80060006

```
(80060006): Index cannot be specified for “operand”.
```

You cannot specify a register number for an operand.

80060007

```
(80060007):Component cannot be specified for “operand”.
```

You cannot specify a component for the operand.

80060009

```
(80060009): '-' cannot be specified for "operand".
```

You cannot specify a minus sign ("-") with the operand.

8006000b

```
(8006000b): Offset index cannot be specified for "operand".
```

You cannot specify an index offset for the operand.

8006000c

```
(8006000c): Address register offset cannot be specified for "operand".
```

You cannot specify an index offset using an address register for the operand.

8006000d

```
(8006000d): Loop counter register offset cannot be specified for "operand".
```

You cannot specify an index offset using the loop-counter register for the operand.

8006000e

```
(8006000e): Loop counter register and address register cannot be specified
together.
```

The loop-counter register and address register cannot be used at the same time.

8006000f

```
(8006000f): Index is not specified in "operand".
```

A register number has not been specified for the operand's register.

80060010

```
(80060010): Invalid index is specified in "operand".
```

The operand's register number exceeds the maximum number of registers.

80060011

```
(80060011): Invalid mask is specified for dest.
```

Masking has been incorrectly specified with the `dest` operand.

Specify masking in x, y, z, w order.

80060012

```
(80060012): Multiple constant registers cannot be specified at the same time.
```

You cannot specify more than one floating-point constant register as an operand at the same time.

80060016

```
(80060016): Src must have one of the following masks: .x|.y|.z|.w.
```

You must use one of the following swizzling specifications with the `src` operand: `.x`, `.y`, `.z`, or `.w`.

80060017

```
(80060017):Src0 and dest cannot be the same.
```

You cannot specify the same register for `src0` and `dest`.

80060018

```
(80060018): Src0 cannot have any swizzle except the default swizzle (.xyzw)
```

You cannot use a swizzling specification other than `.xyzw` with `src0`.

80060019

```
(80060019): Dest must have one of the following masks: .x|.y|.z|.xy|.xz|.yz|.xyz.
```

You must specify one of the following masks with dest: `.x`, `.y`, `.z`, `.xy`, `.xz`, `.yz`, or `.xyz`.

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8006001b

```
(8006001b): Dest must have "mask pattern" mask.
```

You must specify one of the masks shown by the mask pattern with `dest`.

8006001f

```
(8006001f): Dest and src0 cannot be the same.
```

You cannot specify the same register for `dest` and `src0`.

80060020

```
(80060020): Dest and src cannot be the same.
```

You cannot specify the same register for `dest` and `src`.

80060021

```
(80060021): Src0 must have one of the following masks: .x|.y|.z|.w.
```

You must use one of the following swizzling specifications with `src0`: `.x`, `.y`, `.z`, or `.w`.

80060022

```
(80060022): Src1 must have one of the following masks: .x|.y|.z|.w.
```

You must use one of the following swizzling specifications with `src1`: `.x`, `.y`, `.z`, or `.w`.

80060023

```
(80060023): Dest and src1 cannot be the same.
```

You cannot specify the same register for `dest` and `src1`.

80060024

```
(80060024): All operand must be the different register.
```

All operands must be different registers.

80060025

```
(80060025): Dest must have one of the following masks: .x|.y|.xy.
```

You must specify one of the following masks with `dest`: `.x`, `.y`, or `.xy`.

80060026

```
(80060026): Source modifier and swizzling cannot be specified for src1 and src2.
```

You cannot use swizzling or minus signs with `src1` and `src2`.

80060028

```
(80060028): Dest and src1 cannot be the same.
```

You cannot specify the same register for `dest` and `src1`.

80060029

```
(80060029): Src1 cannot have any swizzle except the default swizzle (.xyzw).
```

You cannot use a swizzling specification other than `.xyzw` with `src1`.

8006002a

```
(8006002a): Dest must have one of the following masks: .x|.y|.z|.xy|.xz|.yz|.xyz.
```

You must specify one of the following masks with `dest`: `.x`, `.y`, `.z`, `.xy`, `.xz`, `.yz`, or `.xyz`.

8006002c

```
(8006002c): Source modifier and swizzling cannot be specified for src1.
```

You cannot use swizzling or minus signs with `src1`.

8006002d

```
(8006002d): Source modifier and swizzling cannot be specified for src2.
```

You cannot use swizzling or minus signs with `src2`.

8006002e

```
(8006002e): Invalid index is specified in "operand".
```

An invalid register number has been specified for the operand.

Specify the first register number for the operands of `m4x4` and other instructions. Confirm whether the register numbers used following macro expansion exceed the maximum number of registers.

8006002f

```
(8006002f): Constant register cannot be used for src0.
```


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You cannot use a floating-point constant register for `src0`.

80060030

```
(80060030): Compare mode must be 0 or 1, 2, 3, 4, 5.
```

You must specify a value between 0 and 5 for the comparison mode with the `cmp` instruction.

80060031

```
(80060031): Status register bit must be 0 or 1.
```

You must specify a value of 0 or 1 for the status register.

80060032

```
(80060032): Condition mode is 0:OR 1:AND 2:OnlyStatus0 3:OnlyStatus1.
```

You must specify one of the following conditional modes: 0 (OR), 1 (AND), 2 (OnlyStatus0), or 3 (OnlyStatus1).

80060033

```
(80060033): Address register component must be x or y.
```

You must specify either the x or y component for the address register.

80060036

```
(80060036): Src0 and src1 cannot be the same.
```

You cannot specify the same register for `src0` and `src1`.

80060037

```
(80060037): Src1 and src2 cannot be the same.
```

You cannot specify the same register for `src1` and `src2`.

80060038

```
(80060038): Dest and src2 cannot be the same.
```

You cannot specify the same register for `dest` and `src2`.

8006003b

```
(8006003b): Loop count must be in the range [0, 255].
```

You must use a value between 0 and 255 to set the loop count for the integer register defined by the `defi` instruction.

8006003c

```
(8006003c): Loop counter initial value must be in the range [-128, 127]. The value
in the range of [128, 255] is mapped to [-128, -1].
```

You must use a value between -128 and 127 to set the initial value for the loop-counter register for the integer register defined by the `defi` instruction. To retain compatibility with past versions, you can specify a value in the range of 128 to 255, but these values are the same as -128 to -1.

8006003d

```
(8006003d): Loop counter step must be in the range [-128, 127].
```

You must use a value between -128 and 127 to set the amount by which to increment the loop-counter register for the integer register defined by the `defi` instruction.

80060040

```
(80060040): Multiple input registers cannot be specified at the same time.
```

You cannot specify more than one input register at the same time for the `src` operand. You can specify the same register twice at the same time.

Code 13-1 Example for Error 80060040

```
add  r0 , v0 , v0  // This does not cause an error
add  r0 , v0, v1   // This causes an error
```

13.3 Assembler Warning Messages

This section describes the warning codes and the content of the warnings output by the assembler.

40070001

```
(40070001): Label “label name” is undefined.
```

A label could not be found.

(You can resolve this by linking another object that includes the label.)

40070003

```
(40070003): mova instruction both before else and after endif might cause the
hardware to hang.
```


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The hardware might hang when there is a `mova` instruction both right before an `else` instruction and right after an `endif` instruction. For more information, see section 12.10.1 Invalid Operation Due to a `mova` Instruction.

40070004

```
(40070004): mova instruction at both first and last code in loop statement might
cause the hardware to hang.
```

The hardware might hang when there is a `mova` instruction both right after a `loop` instruction and right before an `endloop` instruction. For more information, see section 12.10.1 Invalid Operation Due to a `mova` Instruction.

400a0001

```
(400a0001): The series of 4 instructions from here might cause the hardware to hang
if this instruction stalls for more than 3 clock cycles.
```

This series of four instructions meets the set of conditions that might cause the hardware to hang. The hardware might hang if the instruction on the indicated line stalls for at least three clock cycles. For more information, see section 12.10.2 Invalid Operation Due to a Specific Order of Instructions.

400a0002

```
(400a0002): The series of 4 instructions from here might cause the hardware to hang
if this instruction stalls for more than 1 clock cycle.
```

This series of four instructions meets the set of conditions that might cause the hardware to hang. The hardware might hang if the instruction on the indicated line stalls for at least two clock cycles. For more information, see section 12.10.2 Invalid Operation Due to a Specific Order of Instructions.

400a0003

```
(400a0003): mova instruction just before end instruction might cause the hardware
to hang.
```

The hardware might hang whenever a `mova` instruction comes just before the `end` instruction. For more information, see section 12.10.1 Invalid Operation Due to a `mova` Instruction.

400a0004

```
(400a0004): mova instruction just before the last instruction writing to output
register might cause the hardware to hang.
```

The hardware might hang whenever a `mova` instruction is followed by an instruction to write to an output register and then by an `end` instruction. For more information, see section 12.10.1 Invalid Operation Due to a `mova` Instruction.

400a0005

```
(400a0005): mova instruction just before branch instruction might cause the
hardware to hang if mova stalled due to a register dependency.
```

The hardware might hang whenever a branch instruction follows a `mova` instruction that depends on the register of a preceding instruction and therefore stalls. For more information, see section 12.10.1 Invalid Operation Due to a `mova` Instruction.

400a0006

```
(400a0006): mova and the following 2 instructions might cause hardware to stop
responding if 1 clock stalling happens right after mova.
```

If the instruction immediately following the `mova` instruction reads a temporary register, and if the `mova` instruction and the following two instructions read an input register, or if any of them reads a temporary register, invalid operations could occur. For more information, see section12.10.1.4 Invalid Operations Caused by Instruction Register Dependencies Before or After a `mova` Instruction.

13.4 Linker Error Messages

This chapter describes the errors and error codes output by the linker.

80080001

```
(80080001): Input file is not specified.
```

An input file has not been specified.

80080005

```
(80080005): “Argument” is not found.
```

The input file could not be found.

80080006

```
(80080006): Exceeded max number of long swizzle masks/patterns.
```

Exceeded the maximum number of swizzling patterns for the `mad` instruction.

80080007

```
(80080007): Exceeded max number of swizzle masks/patterns.
```


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The total number of swizzling patterns has exceeded the limit.

8008000f

```
(8008000f): Label "label name" is duplicate.
```

Duplicate label names have been defined in a subroutine object.

80080012

```
(80080012): Cannot open output file.
```

Cannot generate the executable file.

Confirm whether another file with the same name and read-only or other attributes exists.

80080014

```
(80080014): "Input filename" is invalid file format.
```

The input file is not an object file.

80080015

```
(80080015): Some input files are the same name.
```

Input files with the same name have been specified.

8008001d

```
(8008001d): "Label name" is not subroutine.
```

A `ret` instruction has not been set for a label called as a subroutine from a `call` instruction.

8008001f

```
(8008001f): "Label name" cannot be found in input object files.
```

Could not find a label referenced from an input file.

80080020

```
(80080020): Vertex shader size is over the limit.
```

The maximum number of shader instructions has been exceeded.

You can link a shader with up to 512 instructions.

80080022

```
(80080022): "Register name" is duplicately defined in "object name" and "object
name".
```

A register has been defined with different values in multiple objects using `def`, `defi`, or `defb` instructions.

80080024

```
(80080024): "Register name" is duplicately defined in "object name" and "object
name".
```

`#pragma output_map` definitions have mapped an output register to different output data attributes in multiple objects.

80080025

```
(80080025): symbol "symbol name" is duplicately defined in "object name" and
"object name".
```

`#pragma bind_symbol` definitions have bound a symbol name to different registers in multiple objects.

8008002a

```
(8008002a): symbol "symbol name" in "object name" and "symbol name" in "object
name" are bound to the same register.
```

`#pragma bind_symbol` definitions have bound symbols in two different objects to the same input register.

8008002b

```
(8008002b): "Label name" is duplicately defined in "subroutine object name"
```

A label name in a main object has also been defined in a subroutine object.

8008002c

```
(8008002c): "Output data attribute name" is duplicately defined in "object name"
and "object name".
```

`#pragma output_map` definitions have mapped an output data attribute to different output registers in multiple objects.

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8008002d

```
(8008002d): Main routine cannot be found.
```

The input files do not have an object with the `main` and `endmain` labels.

8008002e

```
(8008002e): Cannot open map file.
```

Cannot generate the map file.

Confirm whether another file with the same name and read-only or other attributes exists.

8008002f

```
(8008002f): No input attribute is defined.
```

No input attributes have been defined.

80080030

```
(80080030): No output map is defined.
```

No output attributes have been defined.

80080031

```
(80080031): -debug and -nodebug cannot be specified together.
```

You cannot specify the `-debug` and `-nodebug` options together.

80080032

```
(80080032): def(bi) in ***.obj and bind_symbol in ***.obj specify the same register
**.
```

You cannot specify the same register with `bind_symbol` and `def` instructions.

80080033

```
(80080033): texture1 and texture2 need to be mapped to same register if 4 textures
are mapped.
```

`texture1` and `texture2` must be mapped to the same register when four textures have been set by `output_map` statements.

80080035

```
(80080035): Invalid warning level is specified
```

An invalid warning level has been specified. You must specify the level as 0, 1, 2, or 3.

13.5 Linker Warning Messages

This section describes the warning codes and the content of the warnings output by the linker.

40090001

```
(40090001): end instruction is not found.
```

The `end` instruction is not found.

40090002

```
(40090002): end instruction is found in loop statement.
```

The `end` instruction was found in the `loop-endloop` statement.

40090003

```
(40090003): end instruction is found in only one of if and else statement.
```

The `end` instruction was found in only one of the if-`else` or `else-endif` statements.

40090004

```
(40090004): input register "Register Name" is not used.
```

There is a possibility that an input register defined by `#pragma bind_symbol` is not used.

40090005

```
(40090005): The access patterns of input registers are different between if and
else statement.
```

The input registers for the `if-else` statement and the `else-endif` statement are different.

40090006

```
(40090006): output register "Register Name" is not set.
```

The output register defined with `#pragma output_map` may not be written.

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40090007

```
(40090007): output register is set in loop statement.
```

The output register is set in the `loop-endloop` instruction.

40090008

```
(40090008): The access patterns of output registers are different between if and
else statement.
```

The output register in use differs between the `if-else` statement and the `else-endif` statement.

40090009

```
(40090009): output register "Register Name" is already set before.
```

The output register is being written to multiple times.

4009000a

```
(40090009a): Recursive call is found, and skipped.
```

A subroutine is being called recursively. The consistency checker feature skips the call that is being called recursively.

4009000b

```
(40090009b): Cannot open file for performance report.
```

Cannot create a file for the output of the result of the performance checker feature.

4009000c

```
(4009000c): mova instruction both before and after returning from subroutine might
cause the hardware to hang.
```

The hardware might hang for a `call`, `callb`, or `callc` code clause when there is a `mova` instruction both right before the final `ret` of the called subroutine and right after the subroutine returns. For more information, see section 12.10.1 Invalid Operation Due to a `mova` Instruction.

4009000d

```
(4009000d): Register "register name" specified as an input source might not be
written before.
```

The register might have been read before it was written to.

14 Warning Levels

Use the `-warn_level=<level>` option in `ctr_VertexShaderLinker32` to specify the warning level. A description of each level is presented below,

- Level 0: No warnings are output.
- Level 1: Standard warnings are output.
- Level 2: In addition to the standard warnings of level 1, warnings are also output when registers are read from before they are written to (that is, when uninitialized registers are read).
- Level 3: In addition to the warnings of level 2 and level 1, warnings are also output by the consistency checker feature.

When the `-warn_level` option has not been specified, the linker runs at warning level 2 by default. An error occurs if a level other than 0–3 is specified

When the warning level is 2 or 3, the dependencies of the assembler instructions are checked in order from the start, with the result that linking takes longer.

For more information about the level 2 warnings output when uninitialized registers have been read, see Section 16.2 Uninitialized Register Checker Feature.

For more information about the level 3 warnings, see Section 16.1 Consistency Checker Feature.

The consistency checker feature's warnings that are output as level 3 warnings were output by the `-check_consistency` option in previous versions. That option remains for compatibility reasons, and when you specify the option it has the same result as specifying level 3 warnings.

The following table shows which warning messages are output for which warning levels. "Yes" indicates that the warning message is output.

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Table 14-1 Warning Messages Output for Each Warning Level

	Warning Level		
Warning code	1	2 (default)	3
40090001			Yes
40090002			Yes
40090003			Yes
40090004			Yes
40090005			Yes
40090006			Yes
40090007			Yes
40090008			Yes
40090009			Yes
4009000a			Yes
4009000b	Yes	Yes	Yes
4009000c	Yes	Yes	Yes
4009000d		Yes	Yes

15 File Format

This chapter describes the format of files generated by the assembler tools.

15.1 Intermediate Object Files

This section describes the format of intermediate object files generated by `ctr_VertexShaderAssembler32.exe`.

15.1.1 Overview

Each file has the following structure.

Figure 15-1 Intermediate Object File Structure

File Header
Setup Information Block
Label Information Block
Program Code Information Block
Swizzle Data Information Block
Line Information Block
Relocation Information Block
Outmap Information Block
Bind Symbol Information Block
String Data Block

The following sections detail each component.

15.1.2 File Header

A fixed file header is placed at the beginning of each file. The header information is used to get the placement and number of data entries for each information block. The file header has the following structure.

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Code 15-1 File Header Structure

```
typedef struct tagOBJFILEHEADER {
 char         signature[4];
 char         version[2];
 unsigned char  shaderType;
 unsigned char  mergeOutputMapsDebug;
 unsigned short inputMask;
 unsigned short outputMask;
 unsigned char  geometryDataMode;
 unsigned char  startIndex;
 unsigned char  variableSizedPrimitiveSize;
 unsigned char  fixedSizedPrimitiveSize;
 unsigned int   setupOffset;
 unsigned int   setupCount;
 unsigned int   labelOffset;
 unsigned int   labelCount;
 unsigned int   instOffset;
 unsigned int   instCount;
 unsigned int   swizzleOffset;
 unsigned int   swizzleCount;
 unsigned int   lineOffset;
 unsigned int   lineCount;
 unsigned int   relocOffset;
 unsigned int   relocCount;
 unsigned int   outmapOffset;
 unsigned int   outmapCount;
 unsigned int   bsymOffset;
 unsigned int   bsymCount;
 unsigned int   stringOffset;
 unsigned int   stringSize;
} OBJFILEHEADER
```

Table 15-1 File Header Fields

Name	Description
`signature`	Stores the string "DVOJ."
`version`	Includes the version of the assembler tool. The first byte is the major version and the second byte is the minor version.
`shaderType`	Set to 0 for a vertex shader object and to 1 for a geometry shader object.
`mergeOutputMapsDebug`	Bit 0 is used by internal settings for the geometry shader. Bit 1 is set equal to 1 for debug builds and to 0 otherwise. Bit 2 contains information about linker warning suppression. Set to 1 if `#pragma suppress_linker_warning_level2_level3` has been defined.
`inputMask`	The input registers information to use. A value of 1 is set for input registers defined by `#pragma bind_symbol`.
`outputMask`	The output registers information to use. A value of 1 is set for output registers defined by `#pragma output_map`.
`geometryDataMode`	Internal information for the geometry shader.
`startIndex`	Internal information for the geometry shader.
`variableSizedPrimitiveSize`	Internal information for the geometry shader.
`fixedSizedPrimitiveSize`	Internal information for the geometry shader.
`setupOffset`	The byte index within the file to the setup information block.
`setupCount`	The number of data entries for setup information.
`labelOffset`	The byte index within the file to the label information block.
`labelCount`	The number of data entries for label information.
`instOffset`	The byte index within the file to the program code information block.
`instCount`	The number of data entries for program code information.
`swizzleOffset`	The byte index within the file to the swizzle data information block.
`swizzleCount`	The number of data entries for swizzle data information.
`lineOffset`	The byte index within the file to the line information block.
`lineCount`	The number of data entries for line information.
`relocOffset`	The byte index within the file to the relocation information block.
`relocCount`	The number of data entries for relocation information.
`outmapOffset`	The byte index within the file to the Outmap information block.
`outmapCount`	The number of data entries for Outmap information.
`bsymOffset`	The byte index within the file to the Bind symbol information block.

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Name	Description
bsymCount	The number of data entries for Bind symbol information.
stringOffset	The byte index within the file to the string data block.
stringSize	The number of bytes in the string data block.

15.1.3 Setup Information

setupCount entries in the setup information block are given by setupOffset in the file header. Setup data is configured by the def, defi and defb instructions in shader assembly. Each setup information entry has the following structure.

Code 15-2 Setup Information Structure

```
typedef struct tagSETUPINFO{
 unsigned short  type;
 unsigned short  index;
 unsigned int   value[4];
} SETUPINFO
```

Table 15-2 Setup Information Fields

Name	Description
type	• 0: Setup information for a Boolean register. • 1: Setup information for an integer register. • 2: Setup information for a floating-point constant register.
index	Register index
value	For a Boolean register, this field has a value of 1 when value[0] is true and 0 when value[0] is false. For an integer register, it stores the three values defined by a defi instruction in bits [7:0], [15:8], and [23:16] of value[0], respectively. For a floating-point constant register, it converts the four values defined by a def instruction into 24-bit floating-point numbers and then stores them in value[0] through value[3].

15.1.4 Label Information

labelCount entries in the label information block are given by labelOffset in the file header. Label information is set in shader assembly. Each label information entry has the following structure.

Code 15-3 Label Information Structure

```
typedef struct tagLABELINFO{
 unsigned int  index;
 unsigned int  address;
 unsigned int  length;
 unsigned int  stringIndex;
} LABELINFO
```

Table 15-3 Label Information Fields

Name	Description
`index`	The label information index. Indexes are numbered starting at `0x00010000` in the order that they were defined in shader assembly.
`address`	The shader program address set by the label.
`length`	The distance from the address set by the label to the `ret` instruction. This is the subroutine length.
`stringIndex`	The byte index in the string data block that stores the label name.

15.1.5 Program Code Information

`instCount` entries in the program code information block are given by `instOffset` in the file header. 32 bits of program code information correspond to a single instruction in shader assembly.

Note: This does not include definition instructions. Macro instructions and control flow instructions may not always have a one-to-one correspondence with program code information.

15.1.6 Swizzle Data Information

`swizzleCount` entries in the swizzle data information block are given by `swizzleOffset` in the file header. Each swizzle data information entry has the following structure.

Code 15-4 Swizzle Data Information Structure

```
typedef struct tagSWIZZLEINFO{
 unsigned int    value;
 unsigned short   usedInfo;
 unsigned short   reserve;
} SWIZZLEINFO
```

Table 15-4 Swizzle Data Information Fields

Name	Description
`value`	The swizzle data itself.
`usedInfo`	Internal information used when linking.
`reserve`	A reserved region.

15.1.7 Line Information

`lineCount` entries in the line information block are indicated by `lineOffset` in the file header. Each line information entry has a one-to-one correspondence with a program code information entry: it stores the filename and line count of the shader object for the program code information entry with the same index. Each line information entry has the following structure.

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Code 15-5 Line Information Structure

```
typedef struct tagLINEINFO{
 unsigned int  stringIndex;
 unsigned int  lineNo;
} LINEINFO
```

Table 15-5 Line Information Fields

Name	Description
`stringIndex`	The index to the region storing the filename of the shader assembly for the corresponding program code. This is the index (in bytes) within the string data block.
`lineNo`	The number of lines of shader assembly for the corresponding program code.

15.1.8 Relocation Information

`relocCount` entries in the relocation information block are indicated by `relocOffset` in the file header. The relocation information is accessed at link time. Each relocation entry has the following structure.

Code 15-6 Relocation Information Structure

```
typedef struct tagRELOCATIONINFO{
 unsigned int   address;
 unsigned short  type;
 unsigned short  reserve;
 unsigned int   stringIndex;
} RELOCATIONINFO
```

Table 15-6 Relocation Information Fields

Name	Description
`address`	The program address associated with the relocation entry.
`type`	• 0: Address relocation • 1: Unresolved subroutine relocation • 4: Swizzle index relocation
`reserve`	A reserved region.
`stringIndex`	The byte offset within the string data block to the location storing the label name for the unresolved subroutine.

15.1.9 Outmap Information

`outmapCount` entries in the outmap information block are indicated by `outmapOffset` in the file header. Outmap information is defined by `#pragma output_map` in shader assembly. Each outmap information entry has the following structure.

Code 15-7 Outmap Information Structure

```
typedef struct tagOUTMAPINFO{
 unsigned short  type;
 unsigned short  index;
 unsigned short  mask;
 unsigned short  reserve;
} OUTMAPINFO
```

Table 15-7 Outmap Information Fields

Name	Description
`type`	This stores the attribute type. • `0: position` • `1: quaternion` • `2: color` • `3: texcoord0` • `4: texcoord0w` • `5: texcoord1` • `6: texcoord2` • `8: view` • `9: generic`
`index`	The output register index.
`mask`	The specified components. These are x, y, z, and w in order from the least-significant bit.
`reserve`	A reserved region.

15.1.10 Bind Symbol Information

`bsymCount` entries in the bind `symbol` information block are indicated by `bsymOffset` in the file header. `Bind symbol` information is defined by `#pragma bind_symbol` in shader assembly. Each bind `symbol` information entry has the following structure.

Code 15-8 Bind Symbol Information Structure

```
typedef struct tagBINDSYMBOLINFO{
 unsigned int   stringIndex;
 unsigned short  startIndex;
 unsigned short  endIndex;
} BINDSYMBOLINFO
```


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Table 15-8 Bind Symbol Information Fields

Name	Description
`stringIndex`	The byte index within the string data block that stores symbol names.
`startIndex`	The starting register index. • 0–15: Input registers 0–15 • 16–111: Floating-point constant registers 0–95 • 112–115: Integer registers 0–3 • 120–135: Boolean registers 0–15
`endIndex`	The ending register index. This has the same values as `startIndex`.

15.1.11 String Data

`stringSize` bytes of string data placed in the string data block are indicated by `stringOffset` in the file header. These include label names, symbol names, filenames, and so on. Each string is delimited by the null character ('`\0`').

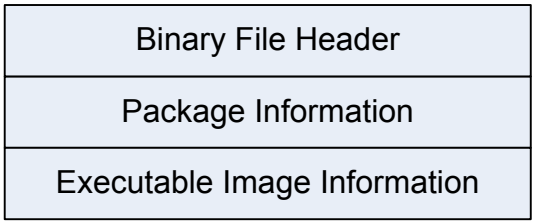
15.2 Executable Binary Files

This section describes the format of executable binary files generated by `ctr_VertexShaderLinker32.exe`.

15.2.1 Overview

Each file has the following structure.

Figure 15-2 Executable Binary File Structure



The following sections give details on each component.

15.2.2 Binary File Header

A variable-length binary file header is placed at the start of the file. The binary file header has the following structure.

Code 15-9 Binary File Header Structure

```
typedef struct tagBINFILEHEADER{
 char         signature[4];
 unsigned int  exeCount;
```

```
 unsigned int  exeOffsetTop;
} BINFILEHEADER
```

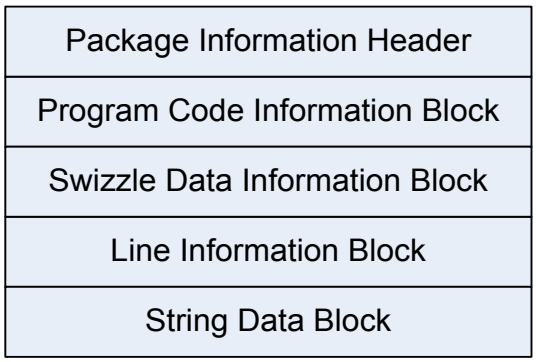
Table 15-9 Binary File Header Fields

Name	Description
`signature`	Stores the string "DVLB."
`exeCount`	The number of information entries for executable images.
`exeOffsetTop`	Stores the byte offset within the file to the first information entry for an executable image. If `exeCount` is 2 or greater, the binary file has multiple executable image information entries. The byte offsets to the executable image information are stored immediately after `exeOffsetTop` using four bytes per entry.

15.2.3 Package Information

Package information is placed immediately after the binary file header. Package information has the following structure.

Figure 15-3 Package Information Structure



The following sections describe each component.

15.2.3.1 Package Information Header

A fixed header is placed at the start of the package information. The placement and number of data entries for each information block is obtained from the header information. The package information header has the following structure.

Code 15-10 Package Information Header Structure

```
typedef struct tagPKGHEADER{
 char         signature[4];
 char         version[2];
 unsigned short reserved0;
 unsigned int   instOffset;
 unsigned int   instCount;
```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```
 unsigned int   swizzleOffset;
 unsigned int   swizzleCount;
 unsigned int   lineOffset;
 unsigned int   lineCount;
 unsigned int   stringOffset;
 unsigned int   stringSize;
} PKGHEADER
```

Table 15-10 Package Information Header Fields

Name	Description
`signature`	Stores the string "DVLP".
`version`	Includes the version of the linker tool. The first byte is the major version and the second byte is the minor version.
`instOffset`	The byte index within the package information to the program code information block.
`instCount`	The number of data entries for program code information.
`swizzleOffset`	The byte index within the package information to the swizzle data information block.
`swizzleCount`	The number of swizzle data information entries.
`lineOffset`	The byte index within the package information to the line information block.
`lineCount`	The number of data entries for line information.
`stringOffset`	The byte index within the package information to the string data block.
`stringSize`	The number of bytes in the string data block.

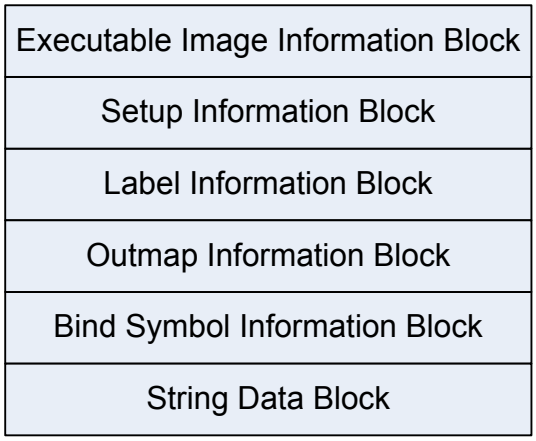
15.2.3.2 Package Information Blocks

Package information and intermediate object file information use the same structure for program code, swizzle data, and line information blocks. For more information, see sections 15.1.5 Program Code Information, 15.1.6 Swizzle Data Information, and 15.1.7 Line Information, respectively. Each package information block that references string data uses an index within the string data block in the package information.

15.2.4 Executable Image Information

Executable image information holds information that is set for each linked object file. Executable image information is generated for each linked main object. Executable image files are placed at the byte offset (within the file) stored in `exeOffsetTop` in the binary file header. When there are multiple executable image information entries, byte offsets to them are stored immediately after `exeOffsetTop` using four bytes apiece. Each information entry for an executable image has the following structure.

Figure 15-4 Executable Image Information Structure



The following sections describe each component.

15.2.4.1 Executable Image Information Header

Each information entry for an executable image has a header. The executable image's information header has the following structure.

Code 15-11 Executable Image Information Header Structure

```
typedef struct tagEXEIMAGEHEADER{
 char         signature[4];
 char         version[2];
 unsigned char  shaderType;
 unsigned char  mergeOutputMapsDebug;
 unsigned int   mainAddr;
 unsigned int   endAddr;
 unsigned short inputMask;
 unsigned short outputMask;
 unsigned char  geometryDataMode;
 unsigned char  startIndex;
 unsigned char  variableSizedPrimitiveSize;
 unsigned char  fixedSizedPrimitiveSize;
 unsigned int   setupOffset;
 unsigned int   setupCount;
 unsigned int   labelOffset;
 unsigned int   labelCount;
 unsigned int   outmapOffset;
 unsigned int   outmapCount;
 unsigned int   bsymOffset;
 unsigned int   bsymCount;
 unsigned int   stringOffset;
 unsigned int   stringSize;
} EXEIMAGEHEADER
```


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Table 15-11 Executable Image Information Header Fields

Name	Description
`signature`	Stores the string "DVLE."
`version`	Includes the version of the assembler tool. The first byte is the major version and the second byte is the minor version.
`shaderType`	Set to 0 for a vertex shader object and to 1 for a geometry shader object.
`mergeOutputMapsDebug`	Bit 0 is used by internal settings for the geometry shader. Bit 1 is set equal to 1 for debug builds and to 0 otherwise. Bit 2 contains information about linker warning suppression. Set to 1 if `#pragma suppress_linker_warning_level2_level3` has been defined.
`mainAddr`	The program address set by the `main` label.
`endAddr`	The program address set by the `endmain` label.
`inputMask`	The input registers information to use. A value of 1 is set for input registers defined by `#pragma bind_symbol`.
`outputMask`	The output registers information to use. A value of 1 is set for output registers defined by `#pragma output_map`.
`geometryDataMode`	Internal information for the geometry shader.
`startIndex`	Internal information for the geometry shader.
`variableSizedPrimitiveSize`	Internal information for the geometry shader.
`fixedSizedPrimitiveSize`	Internal information for the geometry shader.
`setupOffset`	The byte index within the executable image information to the setup information block.
`setupCount`	The number of data entries for setup information.
`labelOffset`	The byte index within the executable image information to the label information block.
`labelCount`	The number of data entries for label information.
`outmapOffset`	The byte index within the executable image information to the Outmap information block.
`outmapCount`	The number of data entries for Outmap information.
`bsymOffset`	The byte index within the executable image information to the Bind symbol information block.
`bsymCount`	The number of data entries for Bind symbol information.
`stringOffset`	The byte index within the executable image information to the string data block.
`stringSize`	The number of bytes in the string data block.

15.2.4.2 Executable Image Information Blocks

Executable image information and intermediate object file information use the same information data structure for setup, label, outmap, and bind `symbol` information blocks. For more information, see sections 15.1.3 Setup Information, 15.1.4 Label Information, 15.1.9 Outmap Information, and 15.1.10 Bind Symbol Information, respectively. Each executable image information block that references string data uses an index within the string data block in the executable image information.

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16 Shader Checking Feature

This section describes the shader checking feature performed by the assembler tool.

16.1 Consistency Checker Feature

The consistency checker feature confirms that the assembly code is properly implemented. This feature is enabled by specifying the `-warn_level=3` option in `ctr_VertexShaderLinker32.exe` and is implemented in the linker. For each main object in the link target, it checks several items. It follows the instructions from the `main` label of the main object and checks up until the `endmain` label. When it finds something it checks for, it outputs a warning.

The consistency checker feature checks each instruction according to the following conditions.

- For `call` instructions, it also checks the call destination.
- It assumes that conditional jump and conditional call instructions do not branch. (The branch destinations for `jpb, jpc, callb`, and `callc` are not considered.)
- For `ifb` and `ifc` instructions, it checks both the `if` items and `else` items.
- Because an infinite loop occurs when a call instruction recursively calls a subroutine, it skips executing the call instructions and goes on to the next instruction when the same subroutine is called more than once by nested `call` instructions. (A warning is output in this case.)

It checks the following items.

- The execution of `end` instructions
- The reading of input registers
- The writing of output registers

The following sections provide details for each item checked. In these sections, the `if` instruction indicates both the `ifb` and `ifc` instructions.

16.1.1 end Instruction Execution Check

Checks whether the `end` instruction was properly called. A warning is output if the following check items are found.

- No `end` instruction is found.
- An `end` instruction is found in a `loop-endloop` statement.
- Only the `if-else` statement or `else-endif` statement has an `end` instruction. If the `if` instruction has no corresponding `else` instruction, and the `end` instruction is found in the `if-endif` statement, a warning is output.

16.1.2 Input Register Read Check

Checks whether the input registers are being used properly. This check targets the input registers and components specified with `#pragma bind_symbol`. A warning is output if the following check items are found.

- When not all input registers are being used, the check reports the names of registers and components that might not be in use.
- When the way in which the input registers are used in the `if-else` and `else-endif` statements differ, this is reported because these differences might be caused by how the `if` instruction branches. If there is no `else` instruction corresponding to the `if` instruction, and an input register is used in the `if-endif` statement, a warning is output.

16.1.3 Output Register Write Check

Checks whether the output registers are being used properly. It targets output registers specified with `#pragma output_map`. In addition, all `xyzw` components are targeted. A warning is output if the following check items are found.

- When not all output registers are written, the check reports the names of registers and components that might not have been written.
- When an output register is set in the `loop-endloop` statement, the check reports this because, depending on how many times the loop statement is repeated, the output register may be written to several times.
- When the way in which the output registers are used in the `if-else` and `else-endif` statements differ, this is reported because these differences might be caused by how the `if` instruction branches. If there is no `else` instruction corresponding to the `if` instruction, and an output register is used in the `if-endif` statement, a warning is output.
- When an output register is being written to multiple times, this check reports the names of the registers or components that might be written multiple times due to writing the instructions in question.

16.2 Uninitialized Register Checker Feature

The uninitialized register checker feature looks at registers that are read and written in the shader and outputs a warning when a register is read before it is written to, because this might result in the referencing of an undefined value. This feature is enabled by specifying the `-warn_level=2` (or `-warn_level=3`) option in `ctr_VertexShaderLinker32.exe` and is implemented in the linker. For each main object in the link target, it checks several items. It follows the instructions from the `main` label of the main object and checks until the `endmain` label, and outputs a warning when it finds something it is checking for.

The feature checks each instruction according to the following conditions.

- For `call` instructions, it checks the call destination.
- It assumes that conditional jump and conditional call instructions do not branch. (The branch destinations for `jpb, jpc, callb`, and `callc` are not considered.)
- `For ifb` and `ifc` instructions, it checks both the `if` item and the `else` item.
- Because an infinite loop occurs when a `call` instruction recursively calls a subroutine, it skips executing the `call` instructions and goes on to the next instruction when the same subroutine is called more than once by nested `call` instructions. (A warning is displayed when this is the case.)

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The following items are checked for:

- Uninitialized temporary registers
- Uninitialized address registers
- Uninitialized loop counter registers
- Uninitialized status registers

The following sections provide details for each item checked. In these sections, the `if` instruction indicates both the `ifb` and `ifc` instructions.

16.2.1 Uninitialized Temporary Register Check

A warning is displayed when a temporary register is read before it is written to. The feature checks each xyzw component. If different temporary registers are written in `if-else` statements and `else-endif` statements, the checker determines they might not be written to.

16.2.2 Uninitialized Address Register Check

A warning is displayed when an address register is read before it is written to. The feature checks each xy component. If different address registers are written in `if-else` statements and `else-endif` statements, the checker determines they might not be written to.

16.2.3 Uninitialized Loop Counter Register Check

A warning is displayed when a loop counter register is read before it is written to. Values are written to loop counter registers by `loop` instructions. If writing or not writing to loop counter registers differs in `if-else` statements and `else-endif` statements, the checker determines they might not be written to.

16.2.4 Uninitialized Status Register Check

A warning is displayed when a status register is read before it is written to. Values are written to status registers by `cmp` instructions. If writing or not writing to status registers differs in `if-else` statements and `else-endif` statements, the checker determines they might not be written to.

16.3 Performance Checker Feature

The performance checker has two features: It estimates the number of clock cycles per vertex it will take during execution based on the implementation of the shader assembly code, and it detects instructions that will generate stalls. The performance checker feature is implemented in the linker, and it is enabled by specifying the `-check_performance` option in `ctr_VertexShaderLinker32.exe`. It checks each main object in the link target. Instructions are tracked from the main label of the main object, and the check is carried out until the `end` instruction or the `endmain` label. The result of the check is output to the command prompt where the linker was executed and at the same time to a file. This output file is created with the same name as the binary file created by the linker, but with the extension changed to ”`perf.txt`”. (It is created in the same location as the binary file.)

The performance checker feature checks each instruction according to the following conditions.

- For `call` instructions, it also checks the call destination.
- It assumes that conditional jump and conditional call instructions do not branch. (The branch destinations for `jpb, jpc, callb`, and `callc` are not considered. The condition is always determined to be `FALSE.`)
- `For ifb` and `ifc` instructions, it only checks the `else` item. (The condition is always determined to be `FALSE.`)
- Because an infinite loop occurs when a `call` instruction recursively calls a subroutine, it skips executing the `call` instructions and goes on to the next instruction when the same subroutine is called more than once by nested `call` instructions.

The following items are output as the result of the check:

- The total number of clock cycles for the executed instructions
- The number of executed instructions
- Instructions that stall, the number of clock cycles, and the cause(s) of the stall.

In calculating the total number of clock cycles, one tick is counted for each instruction, and if there is a stall, the number of ticks of the stall is added to the total. Note that this process does not account for all of the causes of stalls that can take place on the actual hardware. The calculated total number of clock cycles is simply a guideline and does not completely match the total on the actual hardware.

Details of the performance checker feature are presented below.

16.3.1 Detectable Causes of Stalls

This feature can detect a number of the factors that can cause stalls. The following sections detail these factors.

16.3.1.1 Stalls Caused by Instruction Dependencies

The performance checker detects stalls caused by dependent relationships between registers used by instructions. Processing stalls for two instructions called in sequence when the instruction called later references the computation result of the preceding instruction before that computation has been completed, as the later instruction must wait for the computation to complete. The number of clock cycles for this stall depends on the latency of the instruction that was called first and on the number of other instructions that execute between these two instructions. For more information about the latency of various instructions, see section 12.7 Instruction Latency. The registers that are subject to this are the temporary registers and the status registers. For status registers, the `cmp` instructions are sometimes dependent on `ifc`, `callc` and `breakc` instructions.

Code 16-1 Dependency Stall Example

```
dp4  r0 , r1 , r2
add  r3 , r4 , r0
```


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In this case, the `dp4` calculation result written to `r0` is used as a source operand in the very next `add` instruction. The `dp4` instruction has a latency of five clock cycles, so the performance checker detects that execution of the `add` instruction stalls for four clock cycles.

If a particular instruction is dependent on a number of other instructions, the performance checker uses the instruction that stalls for the largest number of clock cycles when calculating the stall duration of that instruction.

Code 16-2 Multiple Dependency Stall Example

```
dp4  r0.x , r1 , r2
mov  r0.y, c0
add  r3 , r4 , r0
```

The `add` instruction in this example depends on both the `dp4` and `mov` instructions because the `r0` register is the destination operand of both. This `add` instruction stalls for three clock cycles because of `dp4` and stalls for one clock cycle because of `mov`. Three clock cycles is the largest value, so the performance checker result is three clock cycles.

16.3.1.2 Stalls Caused by `mova` Instruction Calls

Calls to the `mova` instruction always stall for three clock cycles.

16.3.1.3 Stalls Caused by Branching

Processing stalls for two clock cycles when the program counter is changed by `+1` or more by a `call` or `if` instruction. When processing returns from the subroutine called by the `call` instruction, there is a stall of two clock cycles for the instruction just before the `ret` instruction.

16.3.2 Stalls With Multiple Causes

If a particular instruction stalled for more than one reason, the stall with the largest number of clock cycles is used when calculating the stall for the instruction.

Code 16-3 Multiple Dependency Stall Example 2

```
main:
call label0   // The call to label0
...
end

label0:
dp4  r0.x , r1, r2
mova a0.x, r0.x
ret
```

In this example, the final `mova` instruction in the `label0` subroutine stalls for four clock cycles due to the dependency on the prior `dp4` instruction, then the `mova` call itself stalls for three clock cycles,

and finally the `call` branch instruction causes a stall of two clock cycles. The largest stall value of four clock cycles is used as the calculated result.

16.3.3 Outputting the Results

The performance checker outputs its results to the command prompt where the linker was executed and also to a file. This output file is created with the same name as the binary file created by the linker, but with the extension changed to `perf.txt`.

The content of this output file looks something like the following example.

Code 16-4 Performance Checker Output

```
Main object: obj\VShader.o

Total executed clock count           14 clock
Total executed instruction count       7 instructions

Detail of stall
==============================================================

VShader.vsh(26): 2 clock stall is caused by branch.

VShader.vsh(36): 4 clock stall is caused.
   |
   +--- 3 clock stall is by mova instruction.
   |
   +--- 2 clock stall is by branch.
   |
   +--- VShader.vsh(35): 4 clock is to wait for this instruction to finish writing
r0.

VShader.vsh(30): 1 clock stall is caused by reading temporary register.
   |
   +--- VShader.vsh(29): To wait for this instruction to finish writing r1.
...
```

`Main object` is the name of the main object that is the target of the performance check. A check result is output for each main object in the link target. `Total executed clock count` is the total number of clock cycles per vertex required for execution. The information under `Detail of stall` shows the location and cause of the stall and the number of clock cycles of the stall. For a stall caused by dependencies, the locations of instructions with dependency relationships and the registers that are the causes are shown. If there are multiple causes of stalls, the number of clock cycles for each cause is shown, together with the number of clock cycles that result from the combination of the causes.

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The check results are output in the order of the execution of the instructions, starting from the `main` label. If an instruction is called multiple times in the same subroutine, the performance check is conducted multiple times on the instruction and the same stall information is output multiple times.

Basically, almost the same information that is output to the file is also output to the command prompt where the linker was executed. (The information about dependencies is shown indented.) If the linker is executed from an environment like Microsoft Visual Studio, you can jump to the source file of the shader assembly code by clicking the output result for causes of stalls in the Output window. This is useful for checking the places where problems arise.

Revision History

Version	Revision Date	Description
4.2	2013/06/14	• Added the `#pragma` directive for `suppress_linker_warning_level2_level3`.
4.1	2013/05/22	• Enhanced the linker warnings, adding a feature to specify warning levels and incorporating the consistency checker feature. • Revised the copyright notice.
4.0	2012/11/19	• Changed the names of the members of the file format structure. `subdivPatchSize` was changed to `variableSizedPrimitiveSize` and `constVertexNumber` was changed to `fixedSizedPrimitiveSize`.
3.9	2012/06/19	• Added section12.10.1.4 Invalid Operations Caused by Instruction Register Dependencies Before or After a **`mova`** Instruction. • Separated the error messages and warning messages.
3.8	2012/02/03	• Fixed typos in the linker error messages. • Revised the copyright notice.
3.7	2011/11/30	• Changed the specification for the `init` argument of the `defi` instruction. • 12.7.1 Arithmetic and cmp Instruction Latency Added an explanation about the number of arithmetic instruction stages. • Revised 12.7.2 Branch Instruction Latency
3.6	2011/10/27	• Added a supplemental description for the `litp` instruction. Revised the latency of the `litp` instruction from 2 to 4. • Revised the copyright notation.
3.5	2011/10/05	• 6.2.1 Per-Register Resources • 8.6.1 bind_symbol (symbol_name , start_index [, end_index]) Revised the number of input registers from 12 to 16.
3.4	2011/07/20	• Overall Fixed typos. • 5.1.1 Options Added an explanation of the `-dependency` option to output assembler file dependencies.
3.3	2011/06/21	• Code 8-12 Packing Multiple Attributes into a Single Output Register Revised the sample code.
3.2	2011/06/14	• 12.7.3 Output Order of Calculation Results Added descriptive text. • 12.7.5 Stalls Due to Conflicts Among Arithmetic Units Added descriptive text. • 12.7.7 Unconditional Stalls Added a section.
3.1	2011/06/06	• Overall Added notes about using the `cmp` instruction to distinguish between ∞, -∞, and `NaN`.

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Version	Revision Date	Description
		• 12.7.3 Output Order of Calculation Results Revised explanations. • 12.7.6 Stalls Due to Instruction Dependencies Revised stall conditions. • 12.8 Results of Exceptional Operations Added descriptions of the `rcp`, `rsq`, `exp`, and `log` instructions, and added additional information to their references. • 12.10.2 Invalid Operation Due to a Specific Order of Instructions Revised invalid operation conditions.
3.0	2011/03/17	• Added a description of the -preprocess execution option. • Added supplementary information about Table 6-1 Register Types. • Revised descriptions of `jpb`, `jpc`, and `breakc` instructions. • Added supplementary information about integer registers in the description of the loop instruction. • Partially revised and added to the error messages.
2.9	2011/03/07	• Added restrictions for `breakc`, `ifc`, and `ifb` instructions.
2.8	2011/01/31	• Made corrections in the code sample for loop instructions (section 7.3). • Revised the maximum number of swizzling and masking patterns for map instructions from 64 to 32. • Made corrections in section 12.5 Instructions That Cannot Be Called Consecutively. • Added section 12.10 Shader Implementations That Cause Invalid Operations. • Added descriptions about swizzling.
2.7	2010/11/17	• Added text about the performance checker feature. • Revised invalid error codes.
2.6	2010/11/05	• Added Chapter 15 Shader Checking Feature.
2.5	2010/09/30	• Revised the reserved words for integer registers.
2.4	2010/09/14	• Revised information about the preprocessor pseudo-instruction `#pragma output_map`. • Added supplementary information to section 12.7.3 Output Order of Calculation Results. • Added supplementary information to section 12.7.5 Stalls Caused by Arithmetic Unit Race Conditions. • Added section 12.7.6 Stalls Caused by Instruction Dependencies.
2.3	2010/08/20	• Added support for UTF8 with a byte order mark as an assembler source format.
2.2	2010/07/07	• Deleted the TEG2 Limitations section.
2.1	2010/06/04	• Added Chapter 14 File Format. • Added preprocessor support for evaluating `#if` and `#elif`. • Added the preprocessor statement `#line`. • (English version only) Fixed typos, standardized terminology, and revised throughout for readability.

Version	Revision Date	Description
2.0	2010/05/11	• Added section 12.8 Results of Exceptional Operations. • Moved conditions for generating NaN output from section 12.9 Limitations Related to Invalid Data Output to section 12.8 Results of Exceptional Operations.
1.9	2010/04/23	• Added section 12.9 TEG2 Limitations. • Revised the format of the definition in section 8.5.1 bind_symbol (symbol_name , start_index [, end_index]) to support an unspecified `end_index`. • Added information to Chapter 11 Map Files. • Added section 12.8 Limitations Related to Invalid Data Output.
1.8	2010/04/02	• Added a description of the limitations on writing to output registers to sections 8.6.2 output_map (data_name , mapped_register) and 12.1 Starting and Ending a Shader.
1.7	2010/03/12	• Revised the descriptions of the range of values for the address register. • Added section 12.7.3 Output Order for Calculation Results. • Added section 12.7.4 Stalls Due To Race Conditions When Outputting Calculation Results. • Added section 12.7.5 Stalls Due To Arithmetic Unit Race Conditions. • Added a specification that prohibits output registers from being overwritten. • Revised the description of the `end` instruction. • Added a restriction that requires input registers to be loaded.
1.6	2010/02/15	• Added support for `#elif` and `#if defined`. • Added support for comments delimited by `/*` and `*/`. • Added information about specifications concerning file paths during debug builds.
1.5	2009/12/25	• Fixed typos.
1.4	2009/11/30	• Revised `bind_symbol` specifications for input registers. • Removed `texture3` from the `output_map` settings. • Revised the error messages. • Renamed tools.
1. 3	2009/10/30	• Explained a limitation that prohibits consecutive calls to the `mova` instruction. • Added a note on jump limitations to the `ret` instruction from `jp`-related instructions. • Revised an error in the description of the `abs` instruction.
1. 2	2009/09/10	• Explained limitations on the number of registers with `#pragma output_map`. • Explained the limitations of `call`-related instructions and control instructions. • Explained the limitations of the `loop` instruction. • Removed output registers from the valid `dest` operands for the `crs` instruction.
1. 1	2009/06/25	• Added a note about the `end` instruction for vertex shaders.
1. 0	2009/04/30	• Initial version.

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